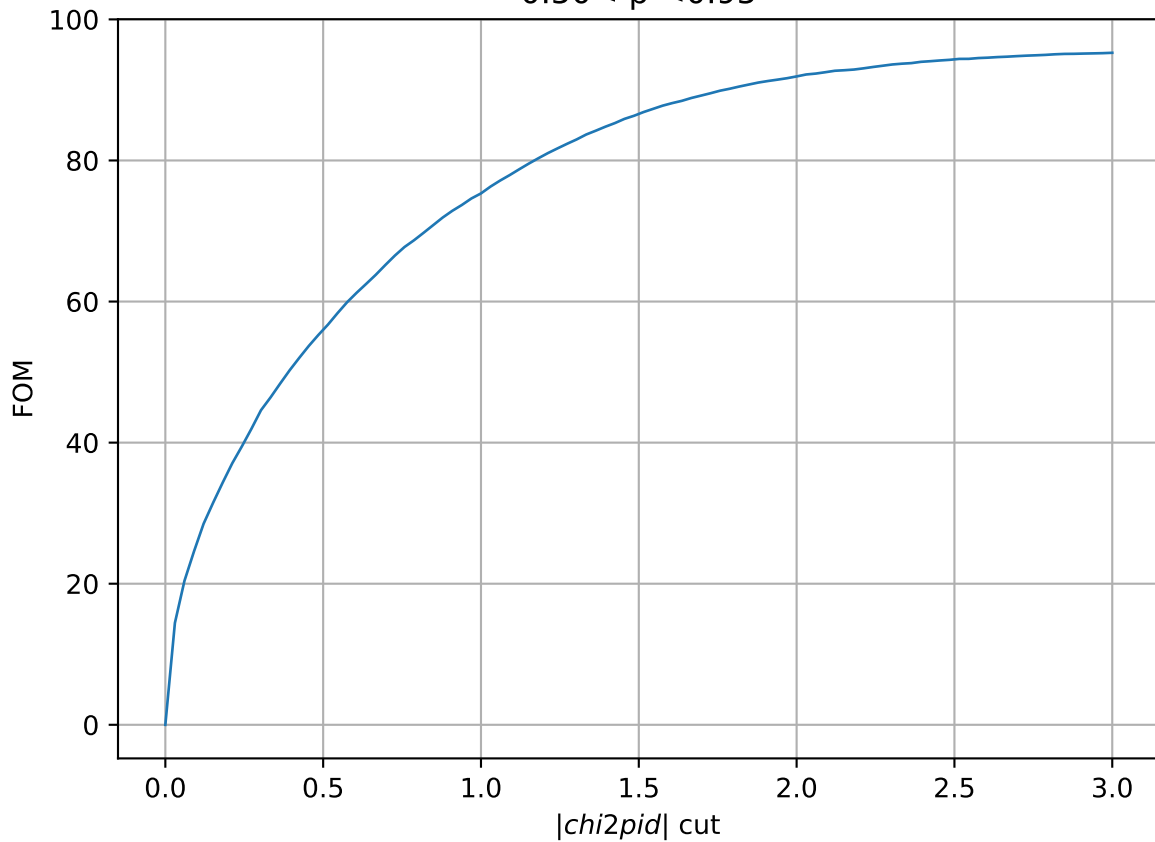
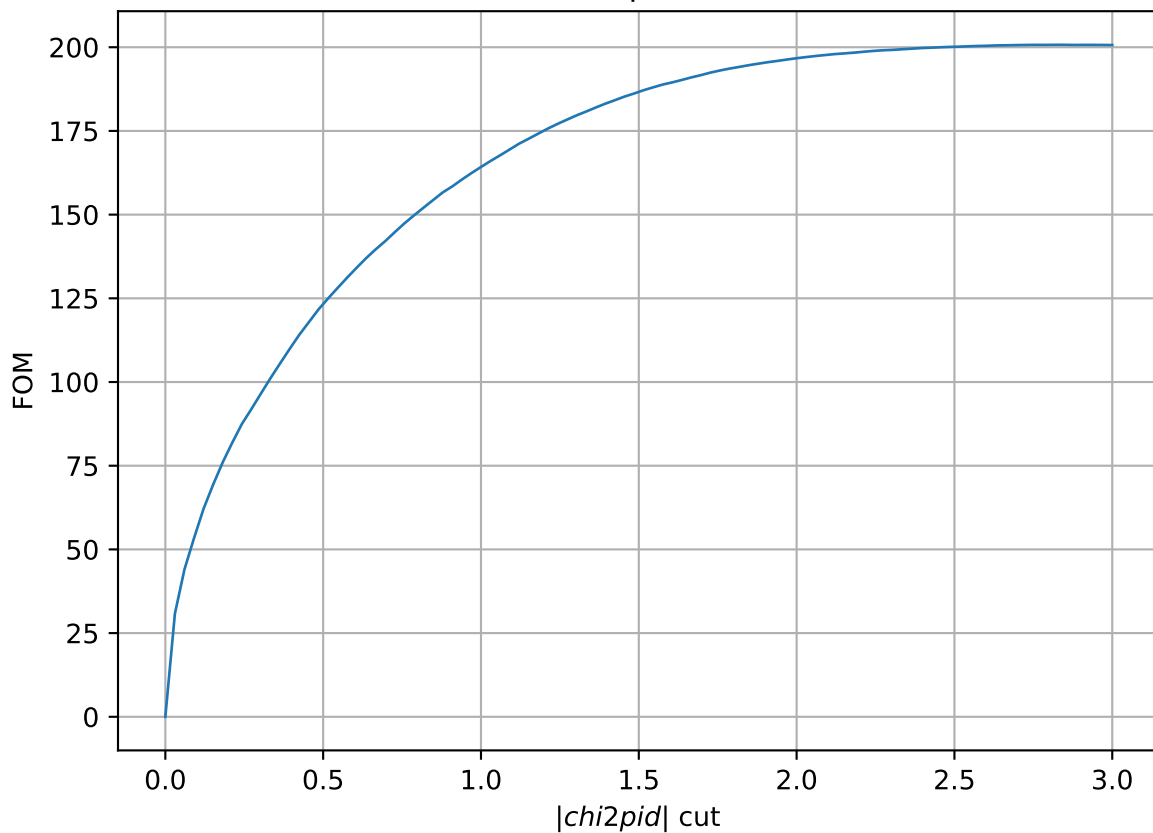


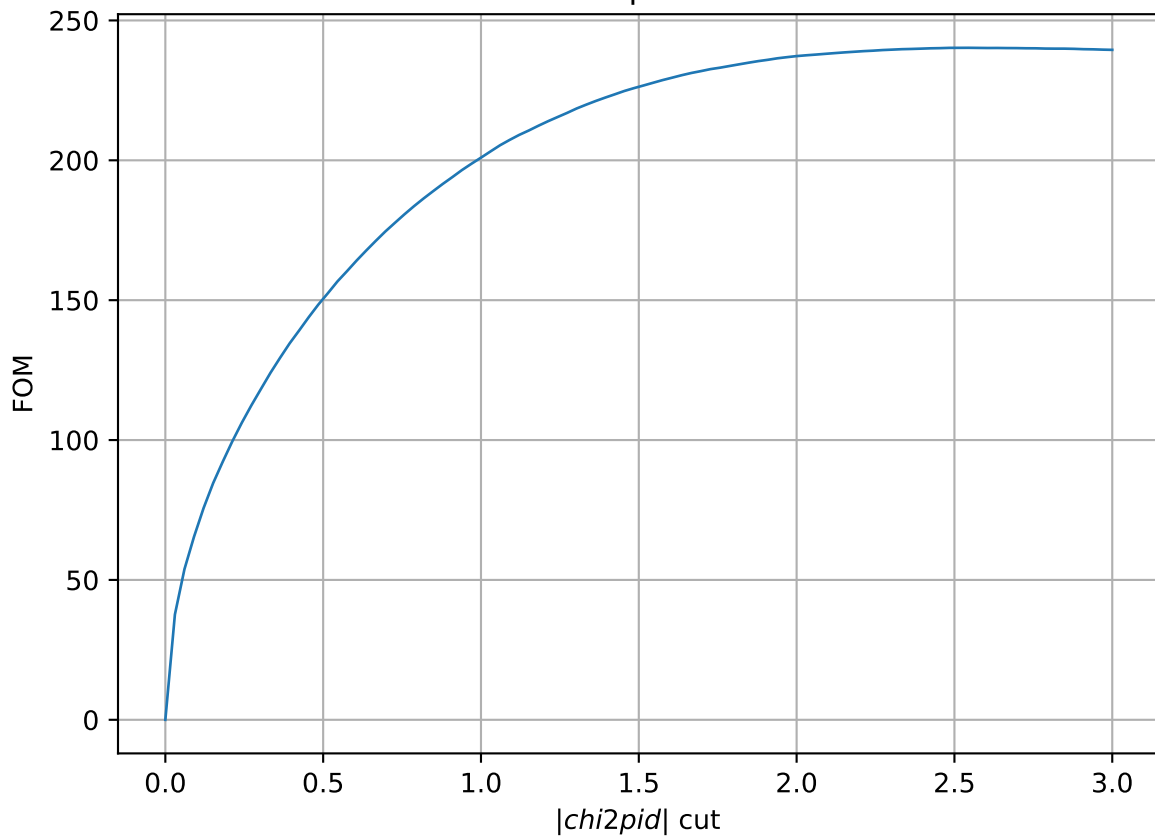
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $0.50 < p < 0.95$



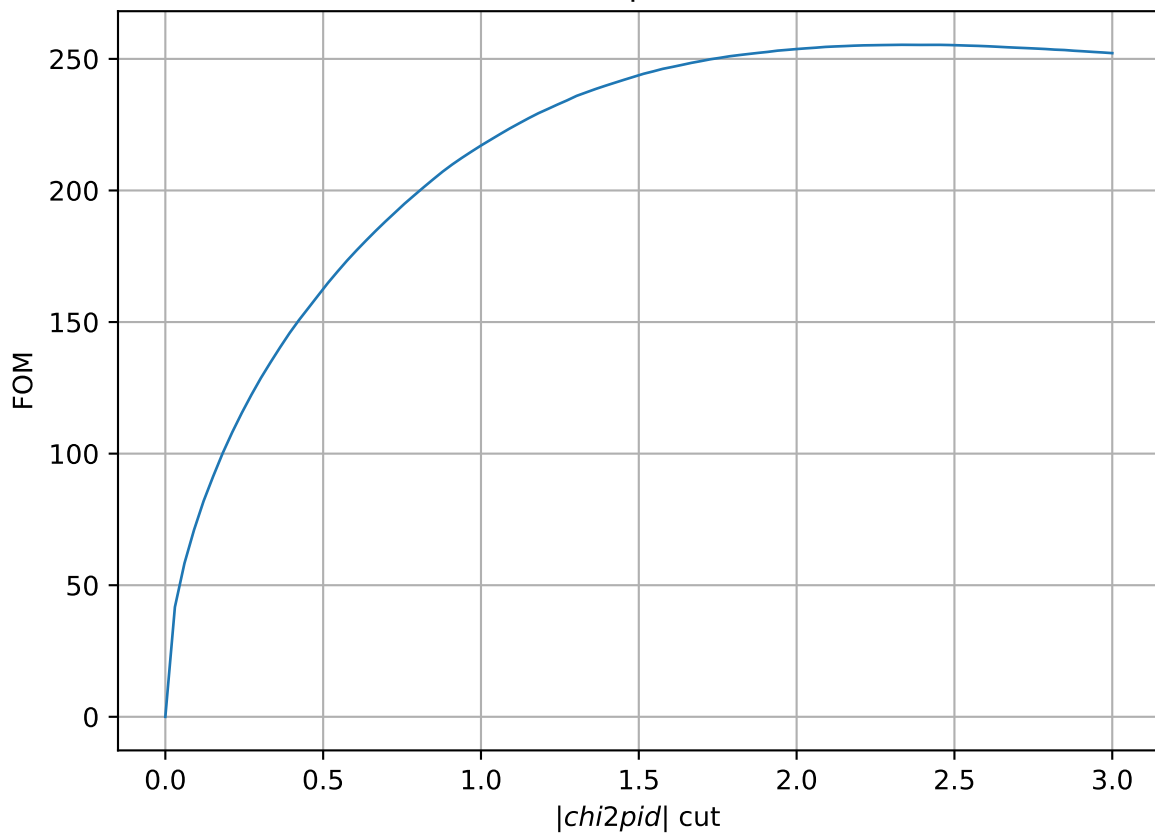
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $0.95 < p < 1.40$



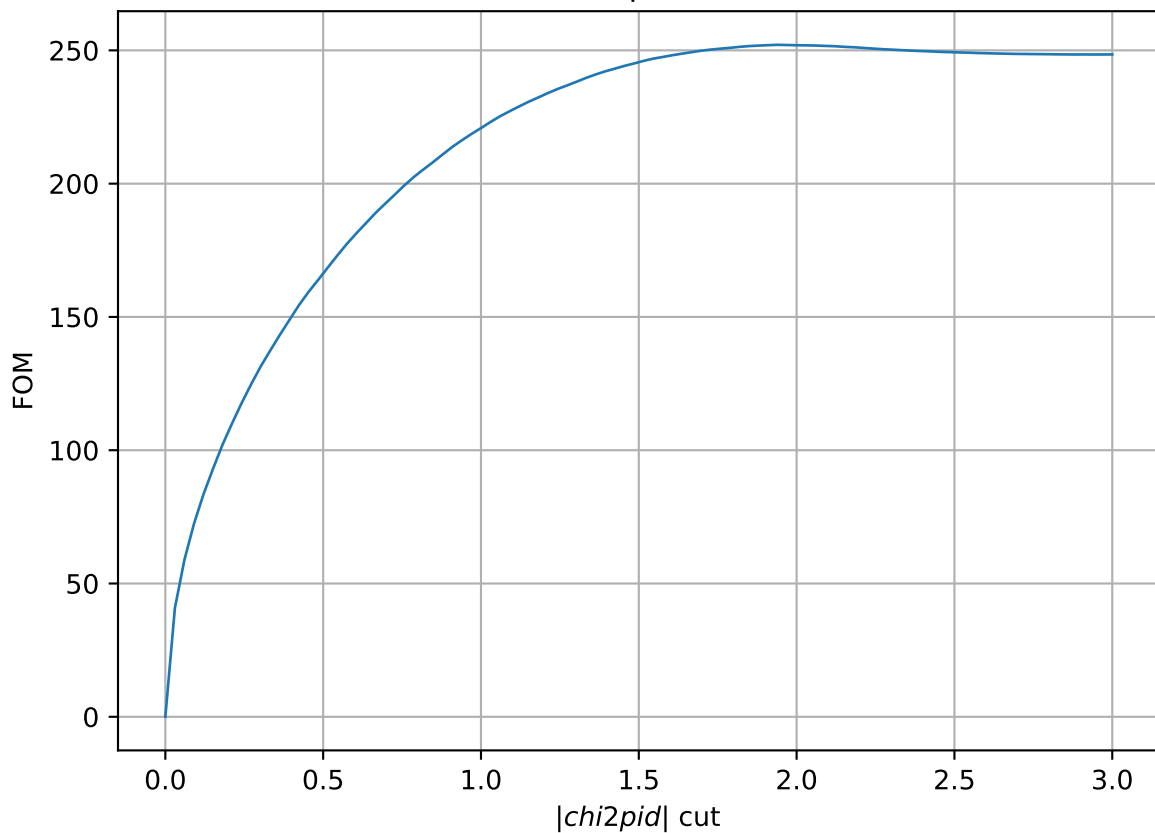
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $1.40 < p < 1.85$



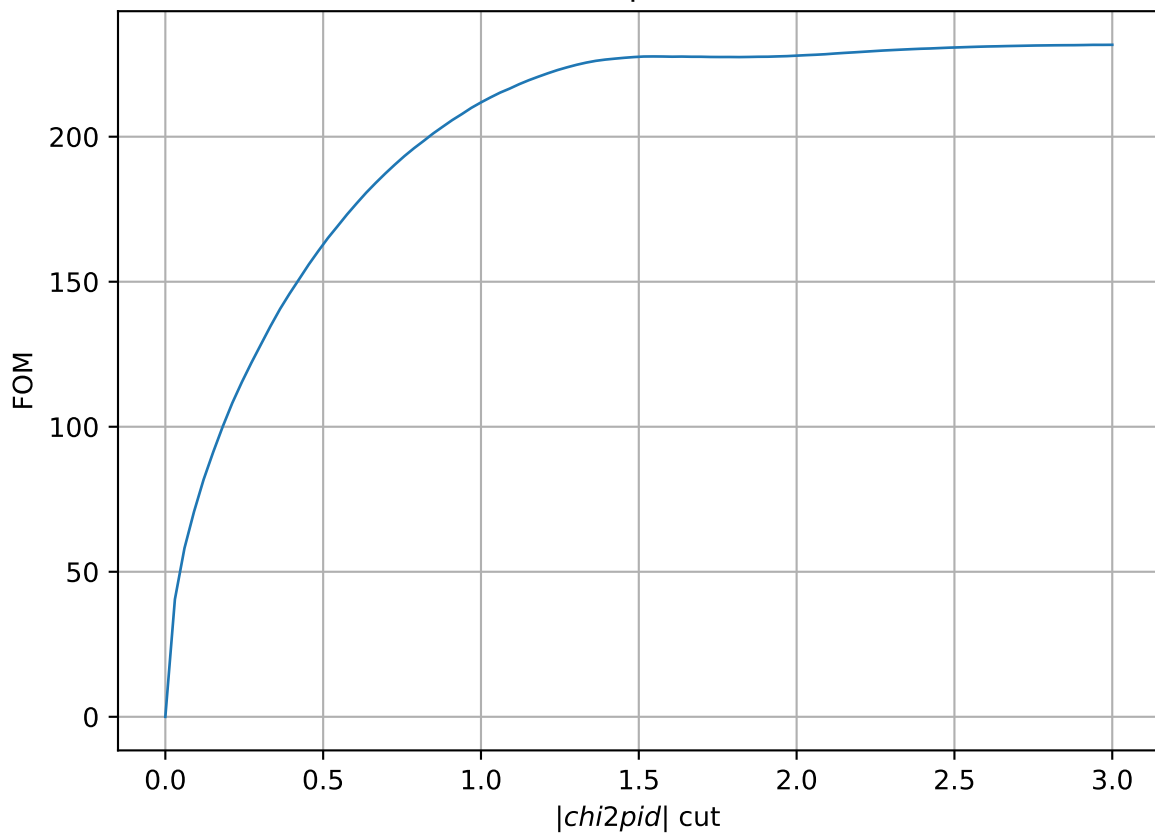
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $1.85 < p < 2.30$



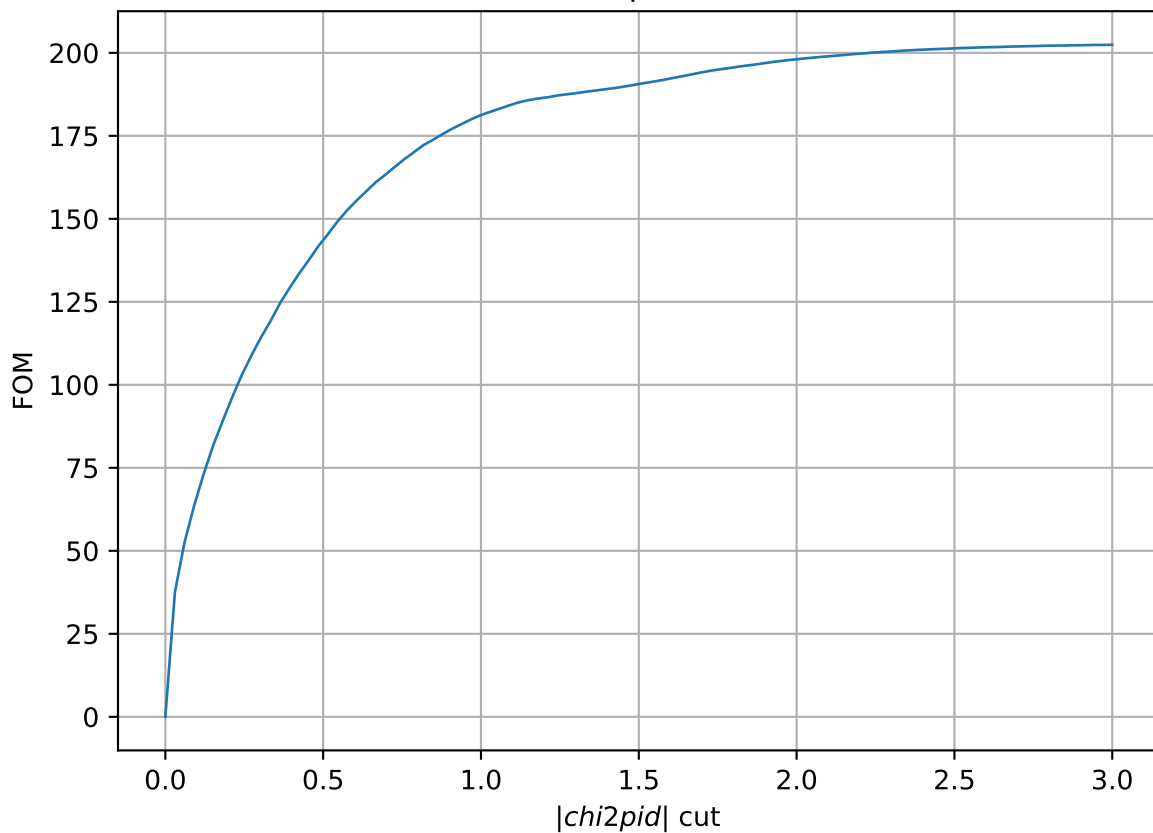
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $2.30 < p < 2.75$



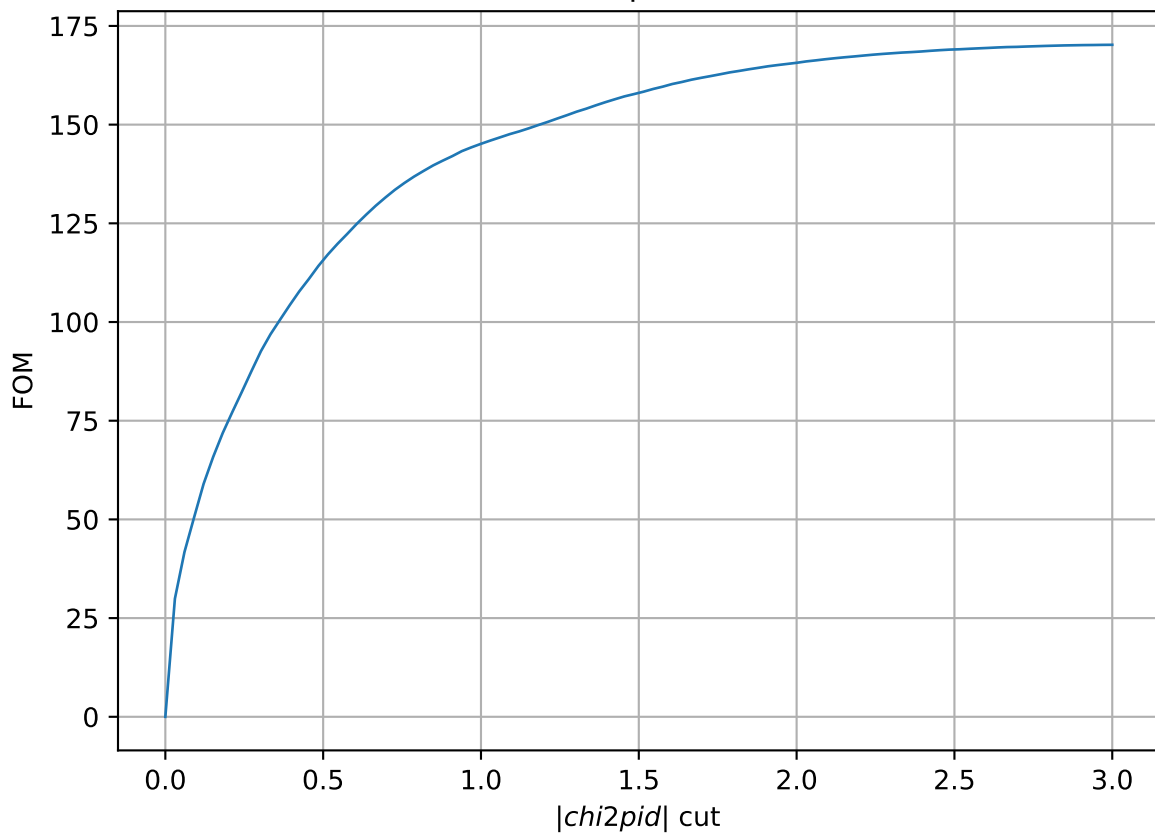
FOM vs $|\chi^2_{pid}|$ cut
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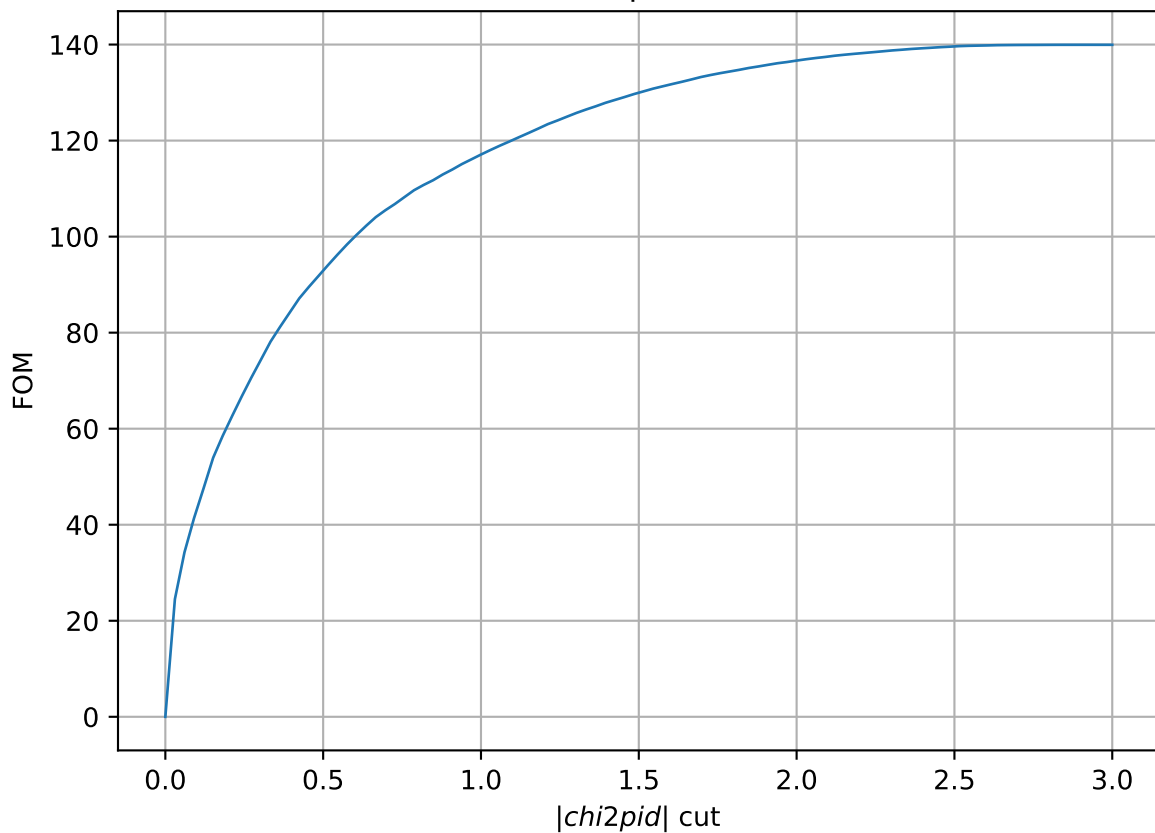
FOM vs $|\chi^2_{pid}|$ cut
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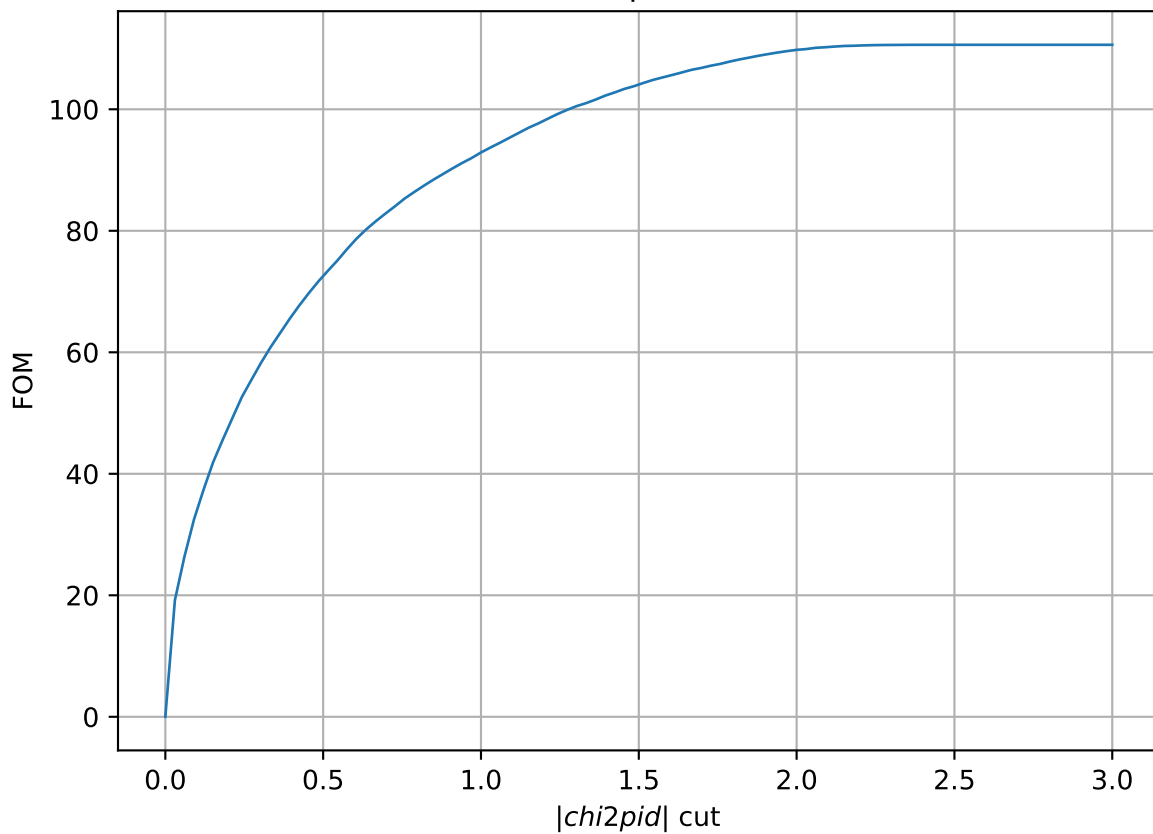
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $3.65 < p < 4.10$



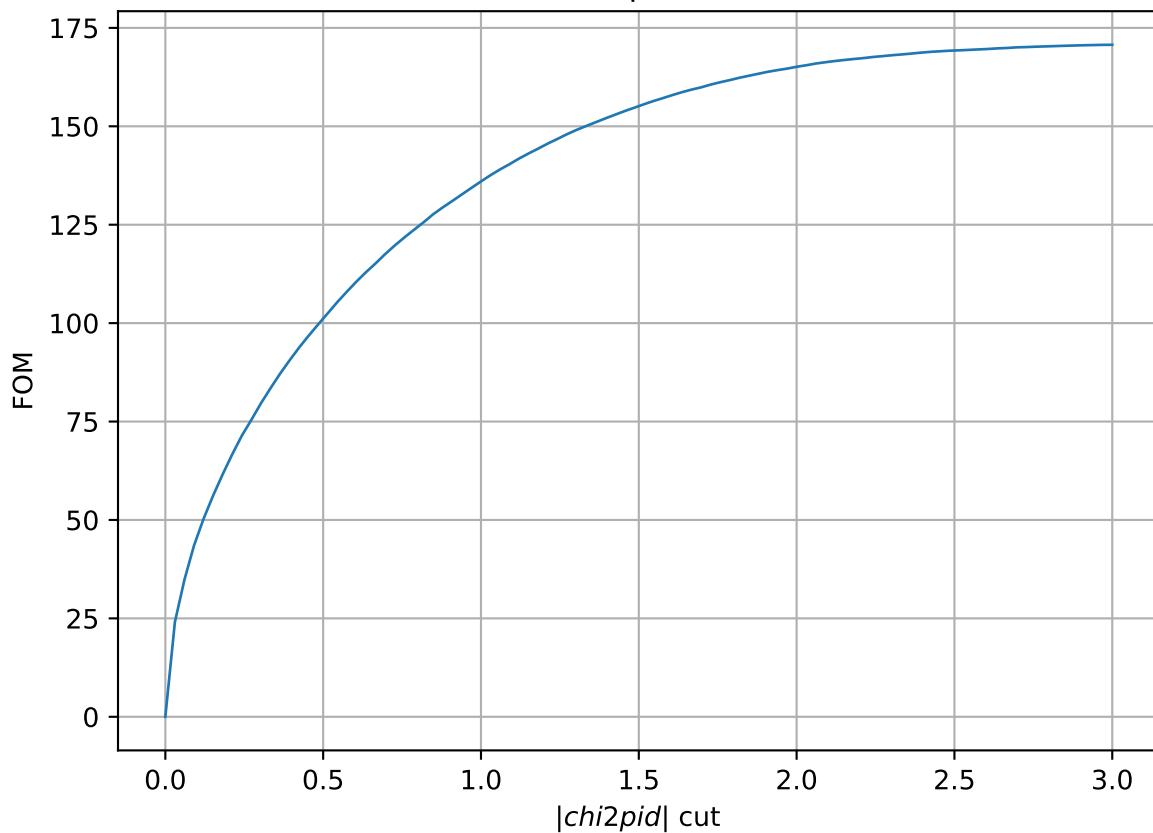
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $4.10 < p < 4.55$



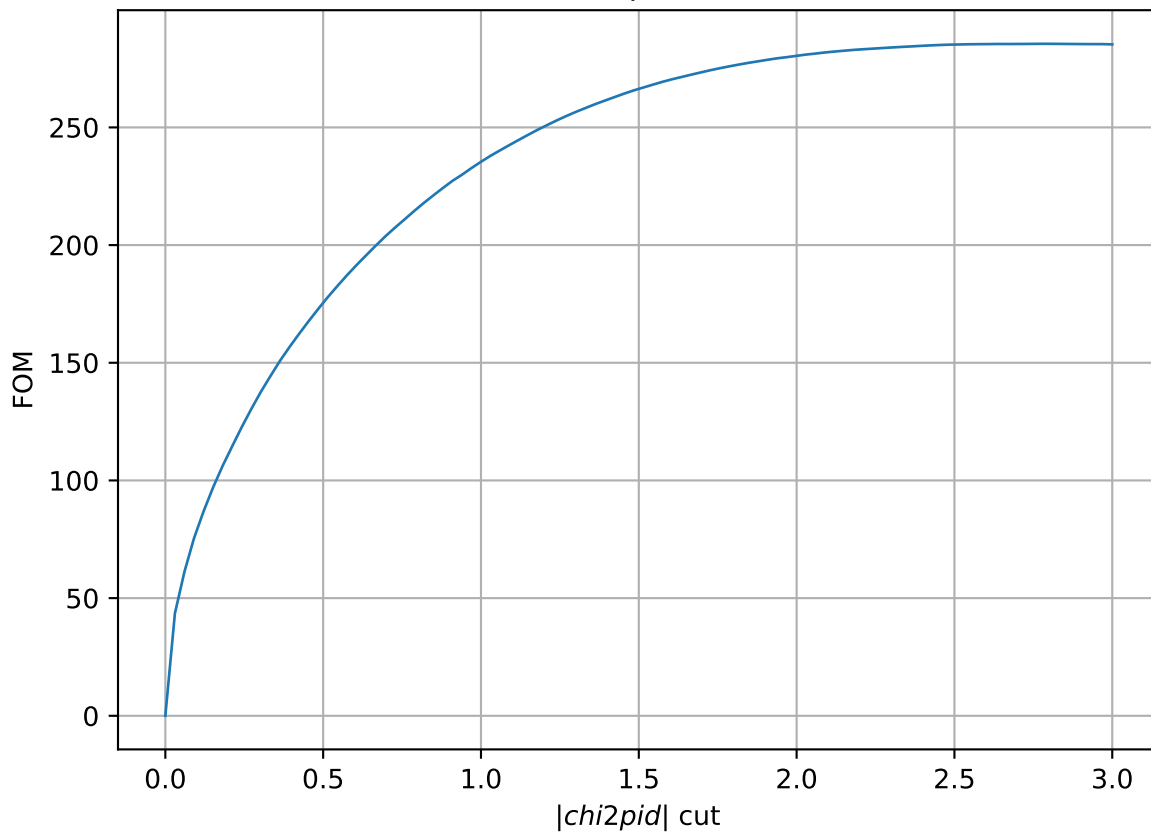
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $4.55 < p < 5.00$



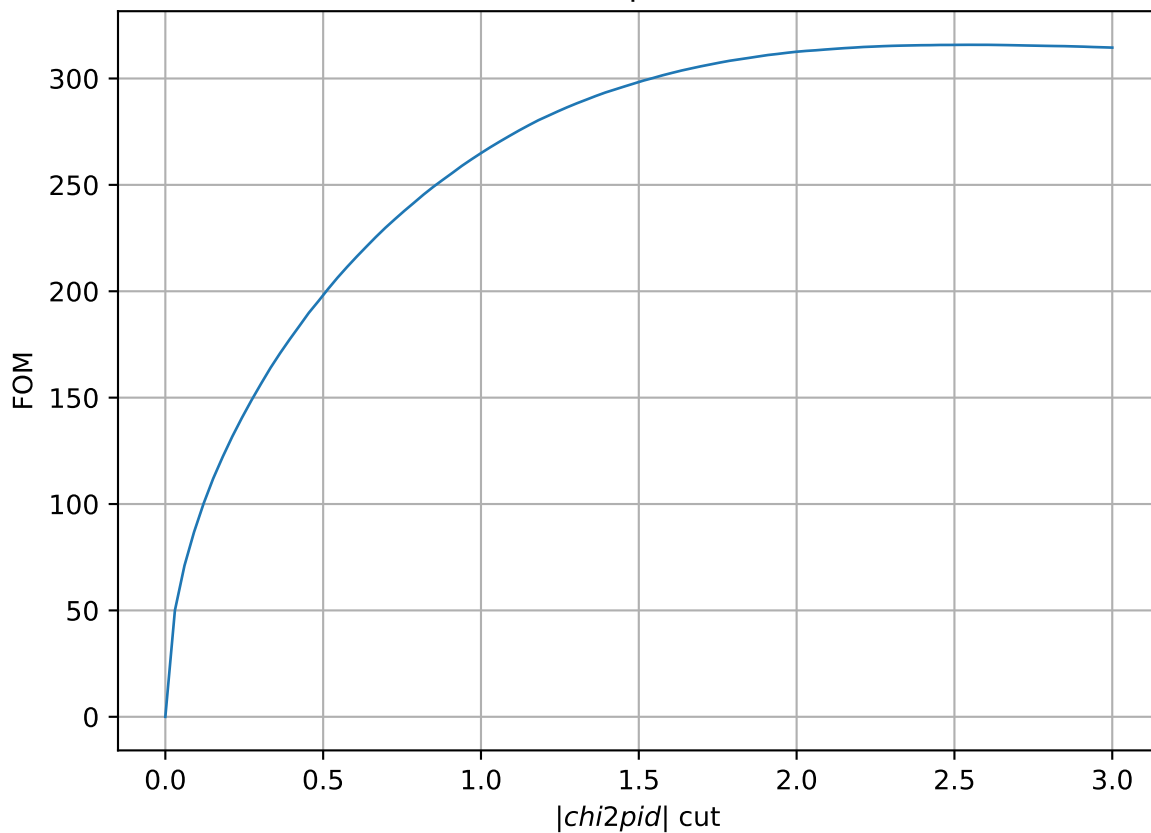
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $0.50 < p < 0.95$



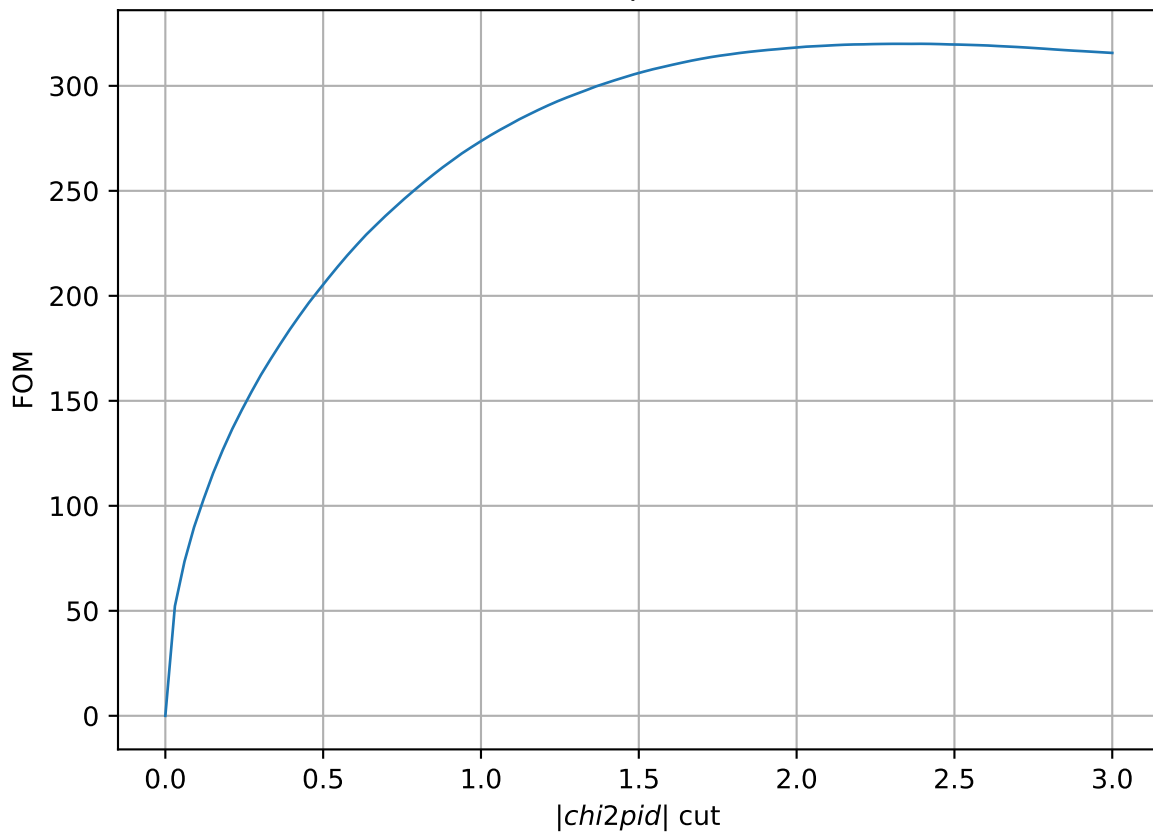
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $0.95 < p < 1.40$



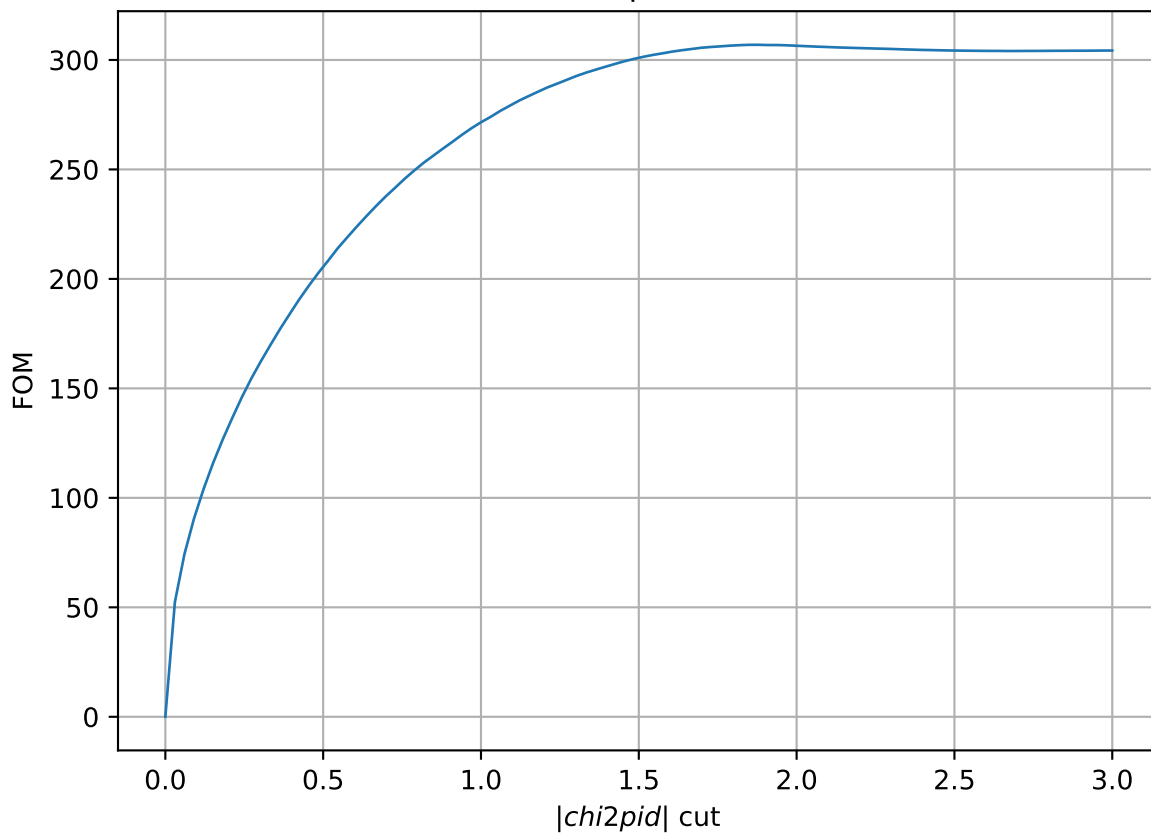
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $1.40 < p < 1.85$



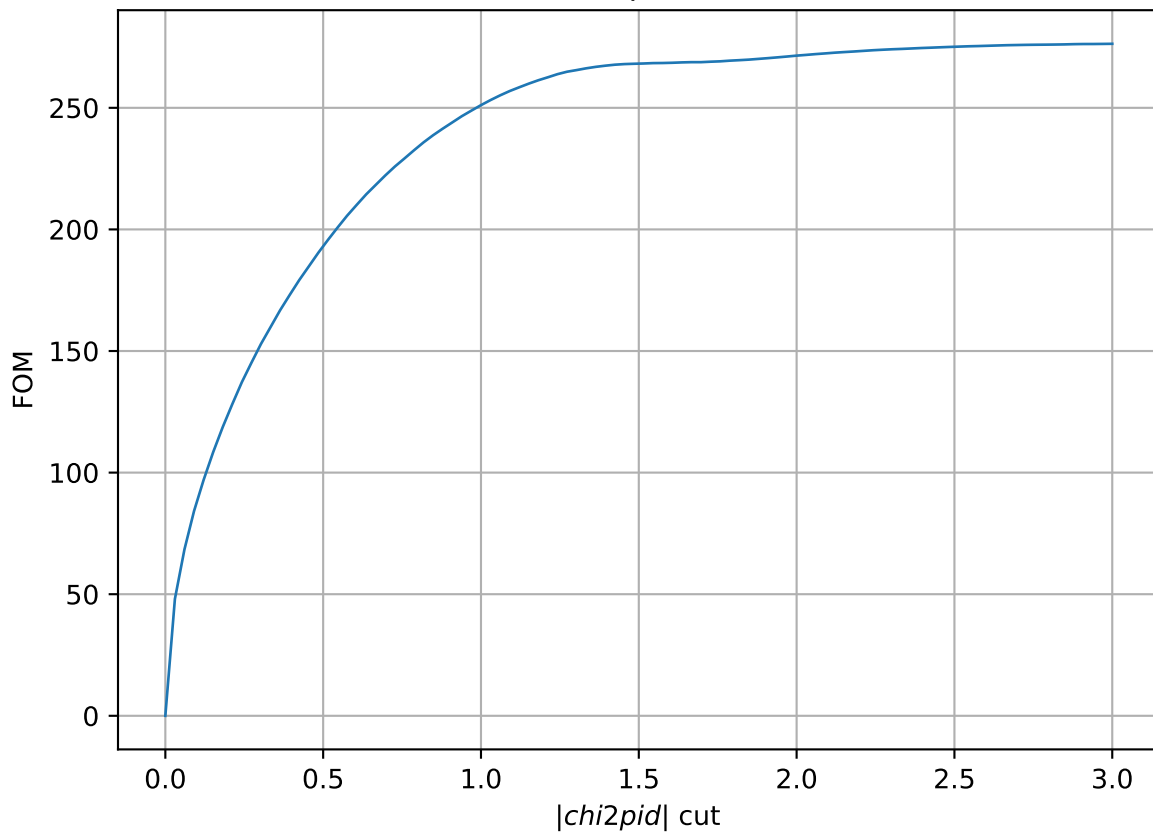
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $1.85 < p < 2.30$



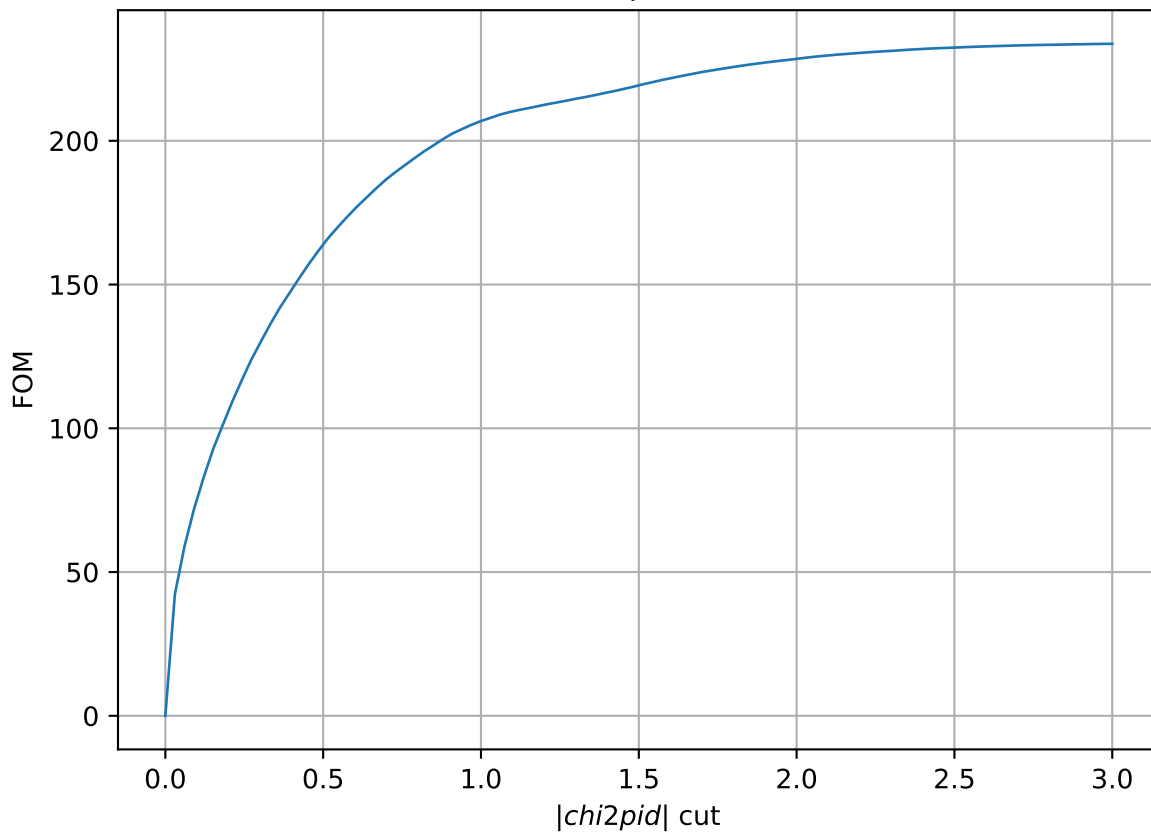
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $2.30 < p < 2.75$



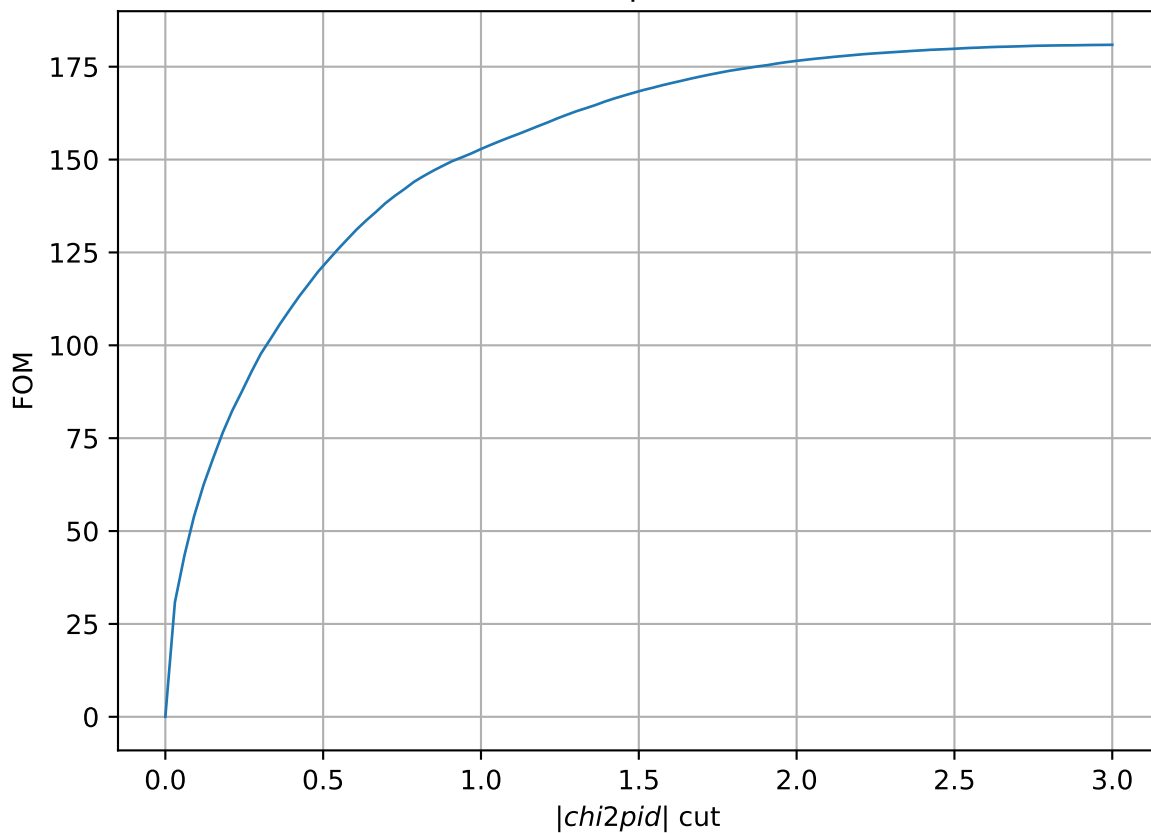
FOM vs $|\chi^2_{pid}|$ cut
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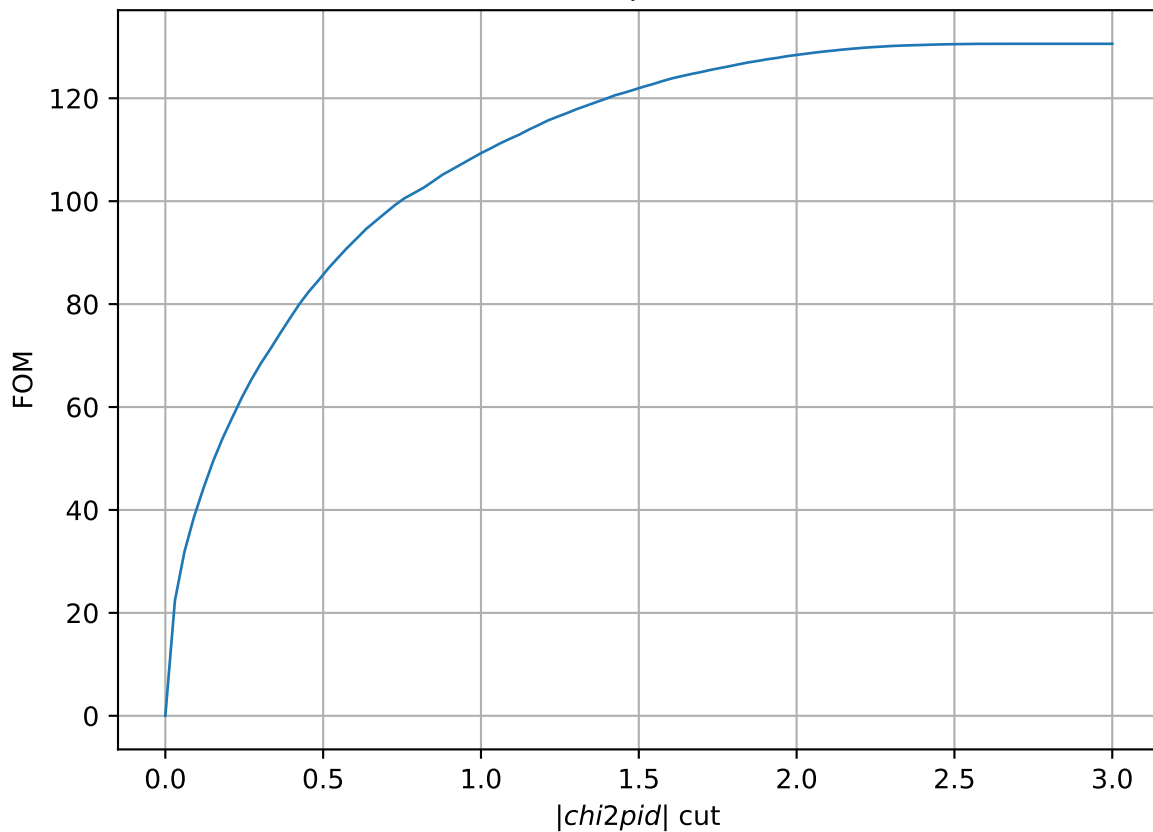
FOM vs $|\chi^2_{pid}|$ cut
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 $3.20 < p < 3.65$



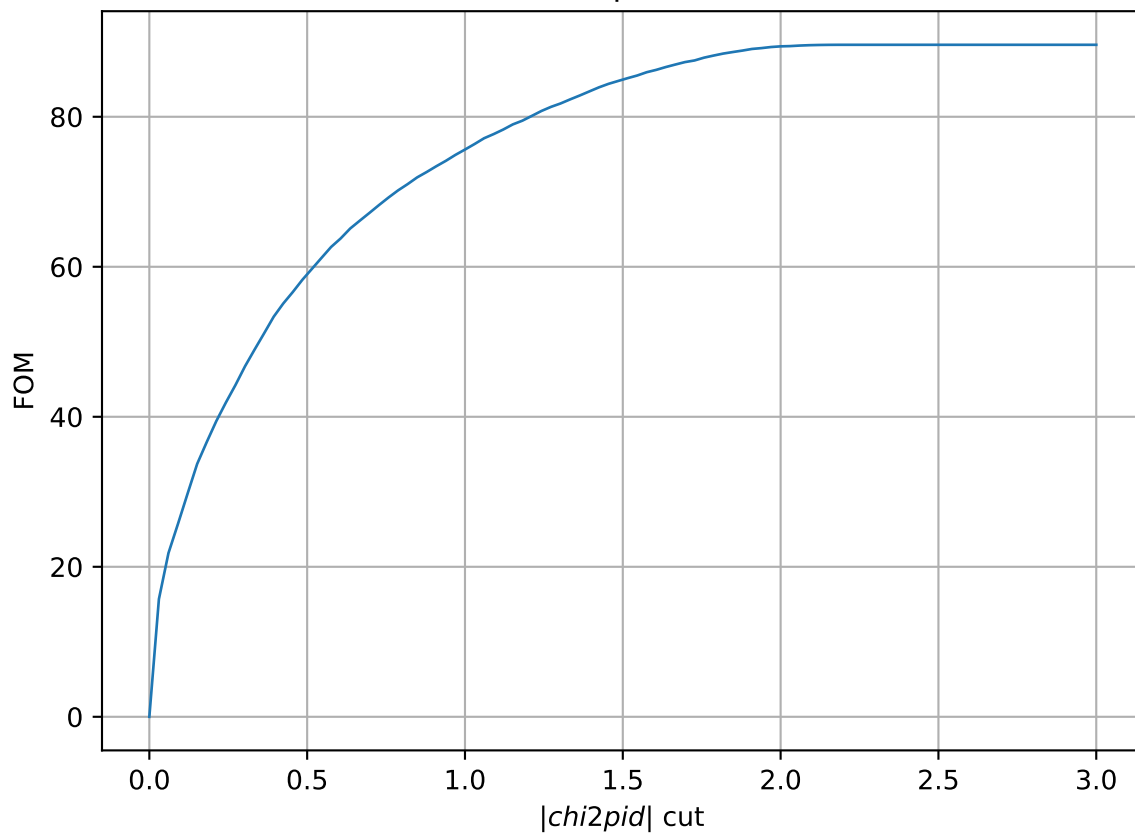
FOM vs $|\chi^2_{pid}|$ cut
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 $3.65 < p < 4.10$



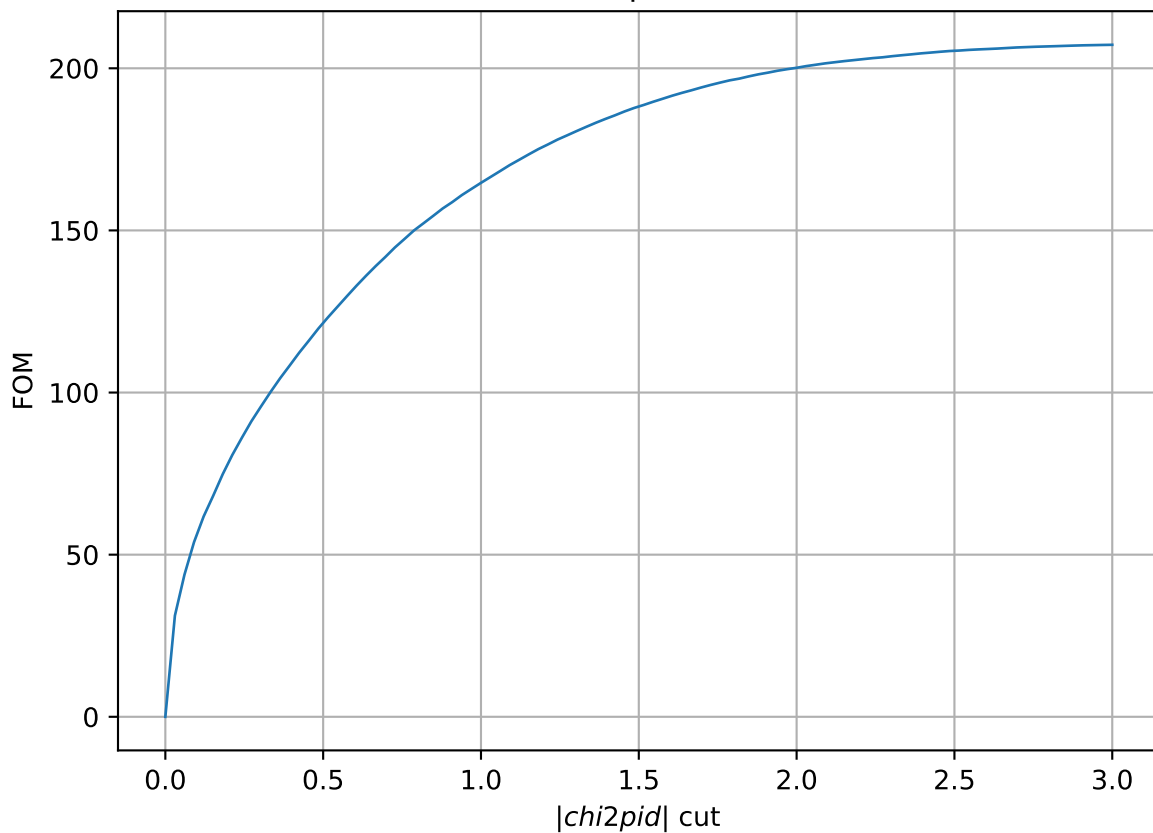
FOM vs $|\chi^2_{pid}|$ cut
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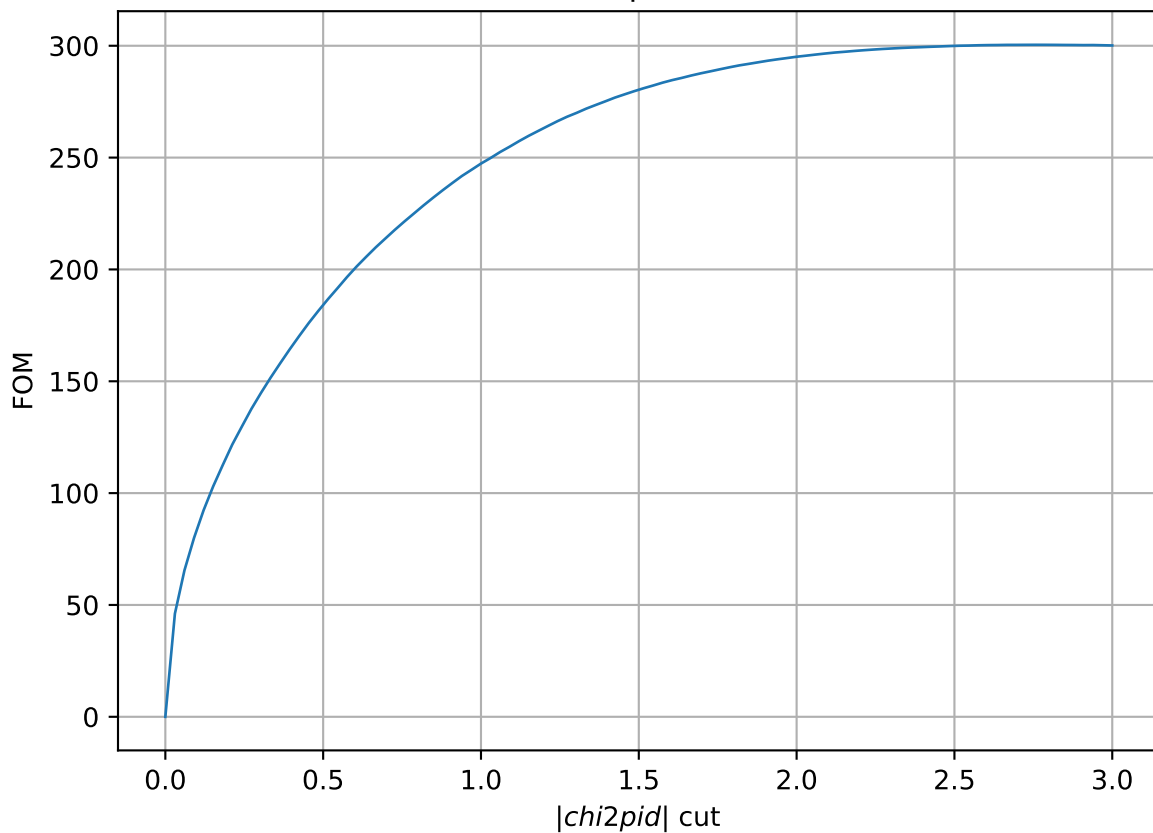
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $4.55 < p < 5.00$



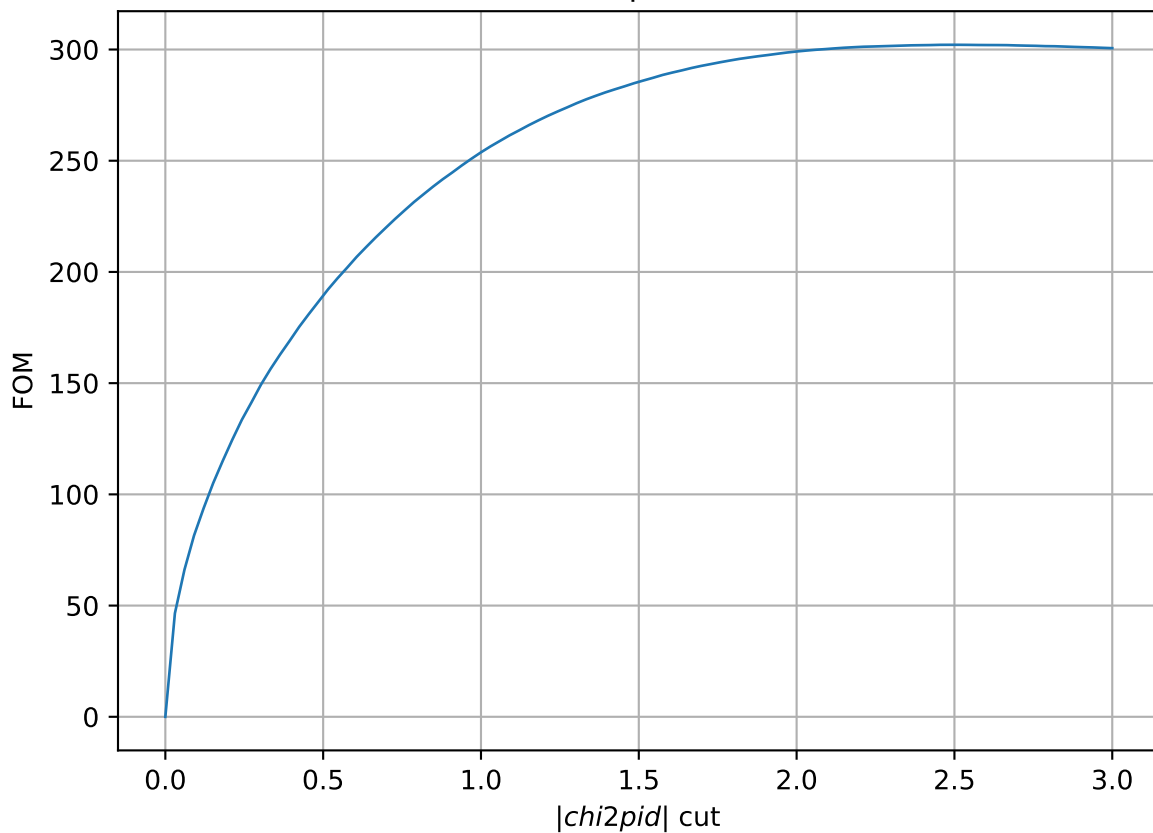
FOM vs $|\chi^2_{pid}|$ cut
 $17.0 < \theta < 23.0$
 $0.50 < p < 0.95$



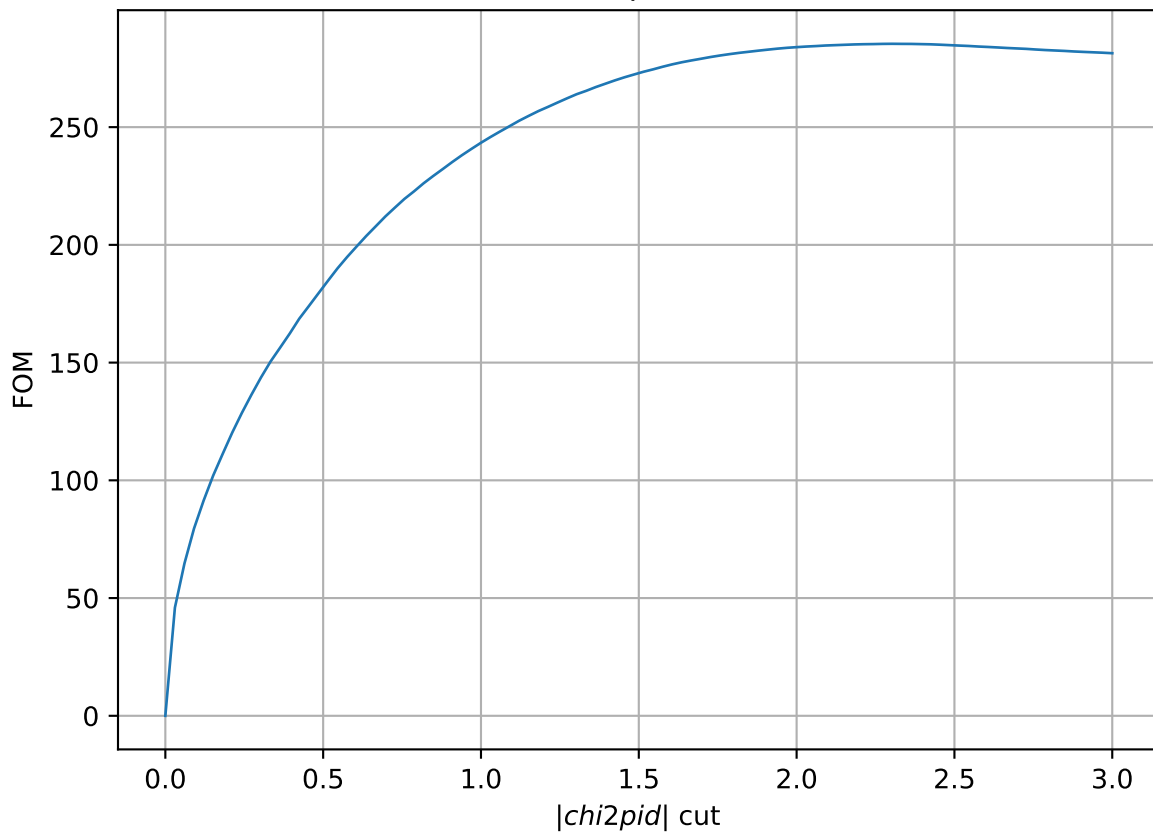
FOM vs $|\chi^2_{pid}|$ cut
 $17.0 < \theta < 23.0$
 $0.95 < p < 1.40$



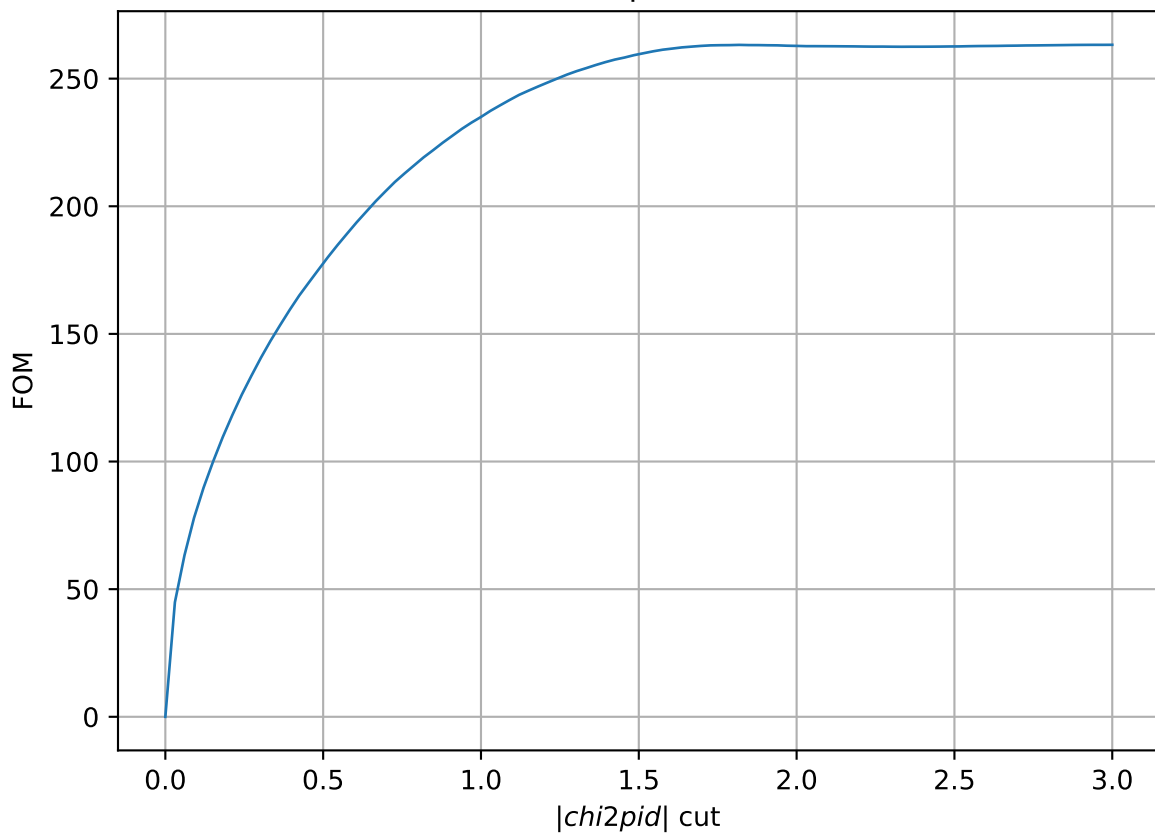
FOM vs $|\chi^2_{pid}|$ cut
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 $1.40 < p < 1.85$



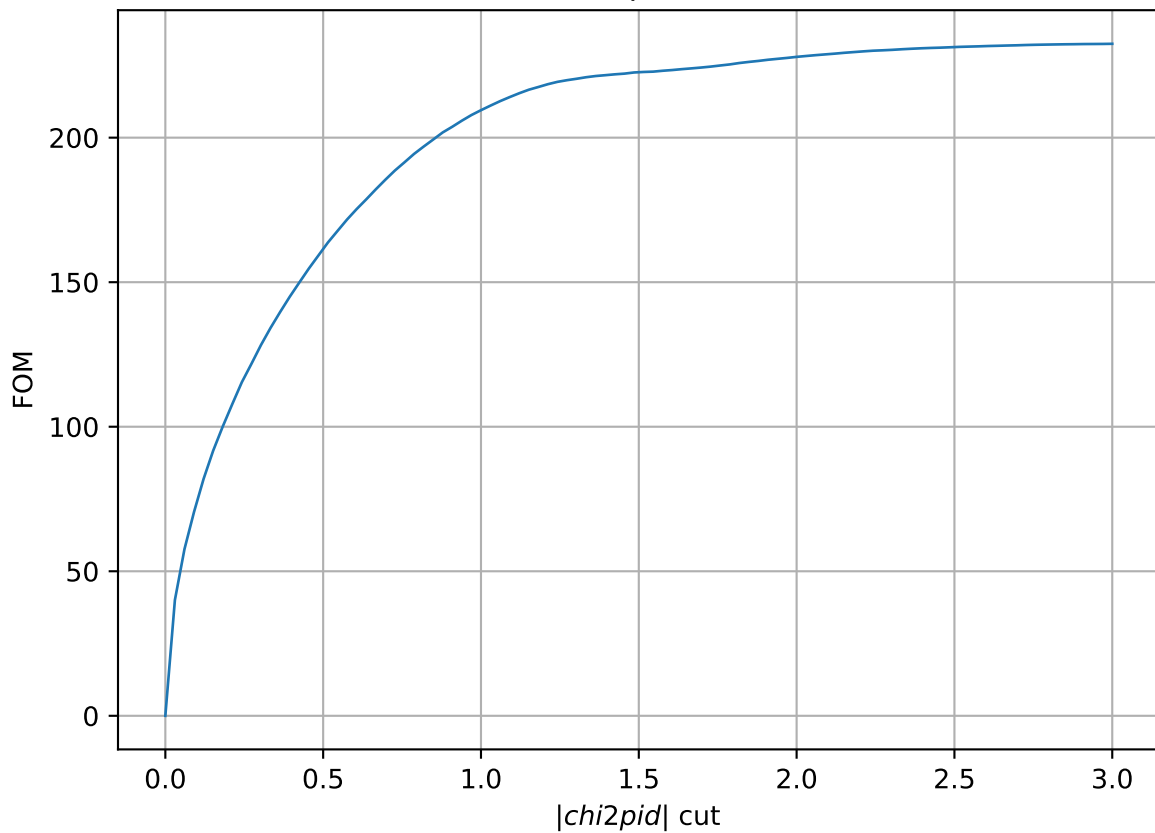
FOM vs $|\chi^2_{pid}|$ cut
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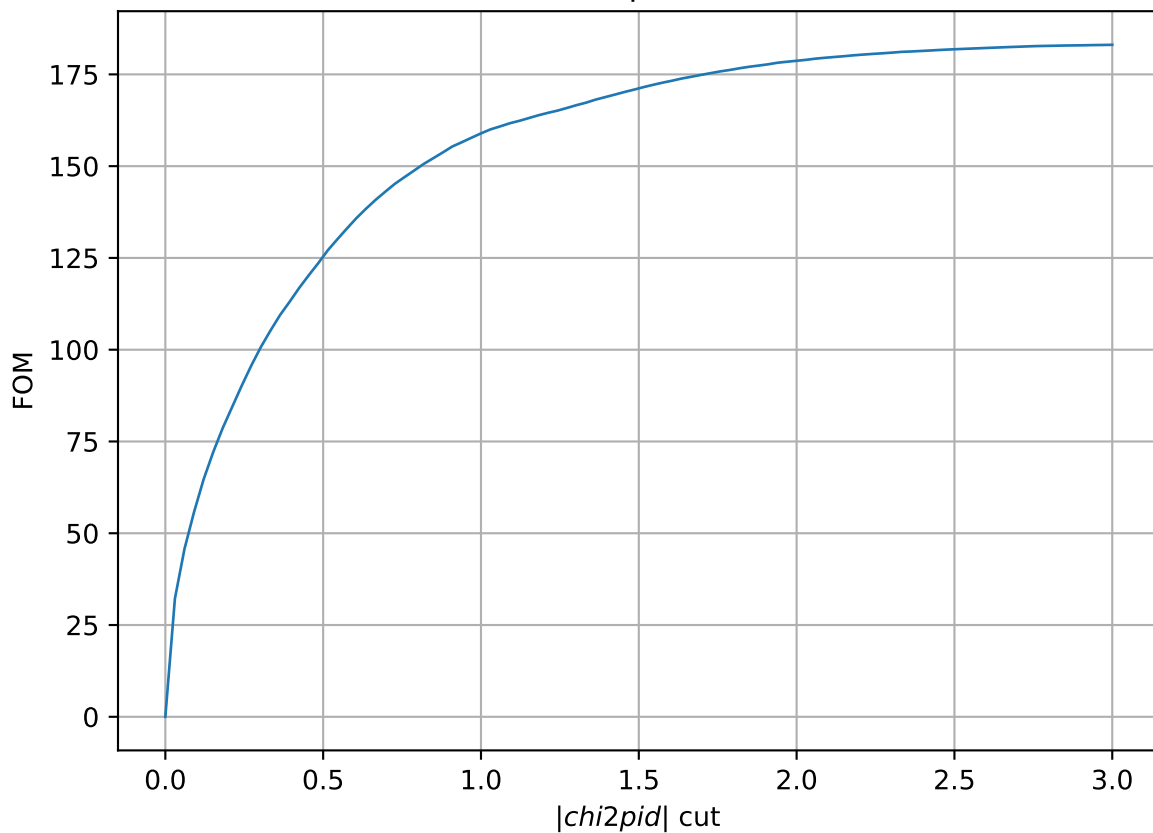
FOM vs $|\chi^2_{pid}|$ cut
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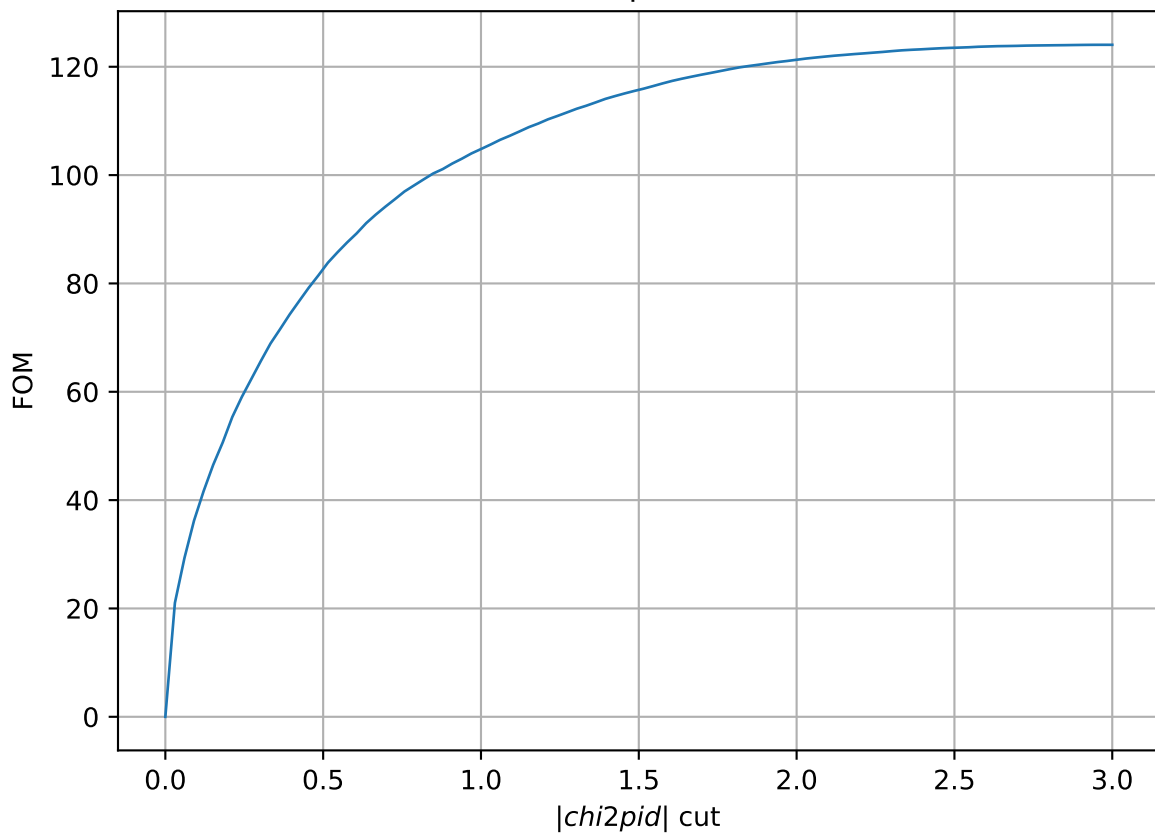
FOM vs $|\chi^2_{pid}|$ cut
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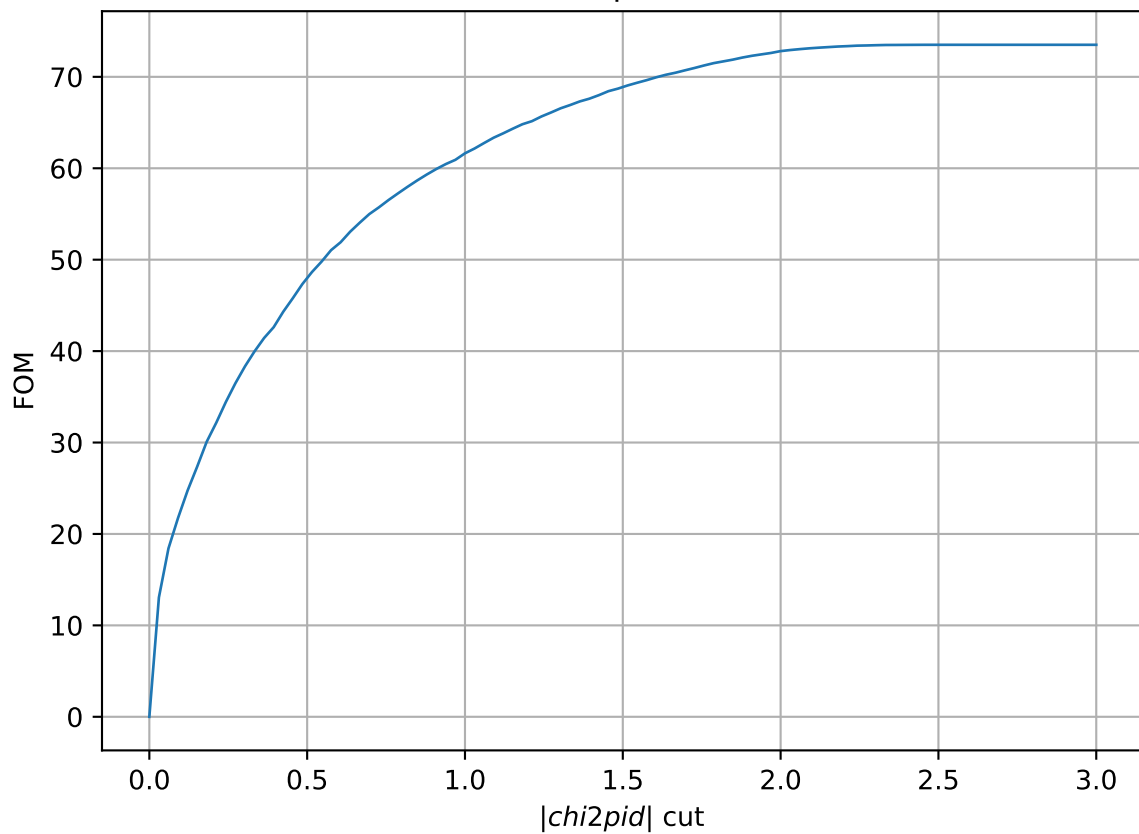
FOM vs $|\chi^2_{pid}|$ cut
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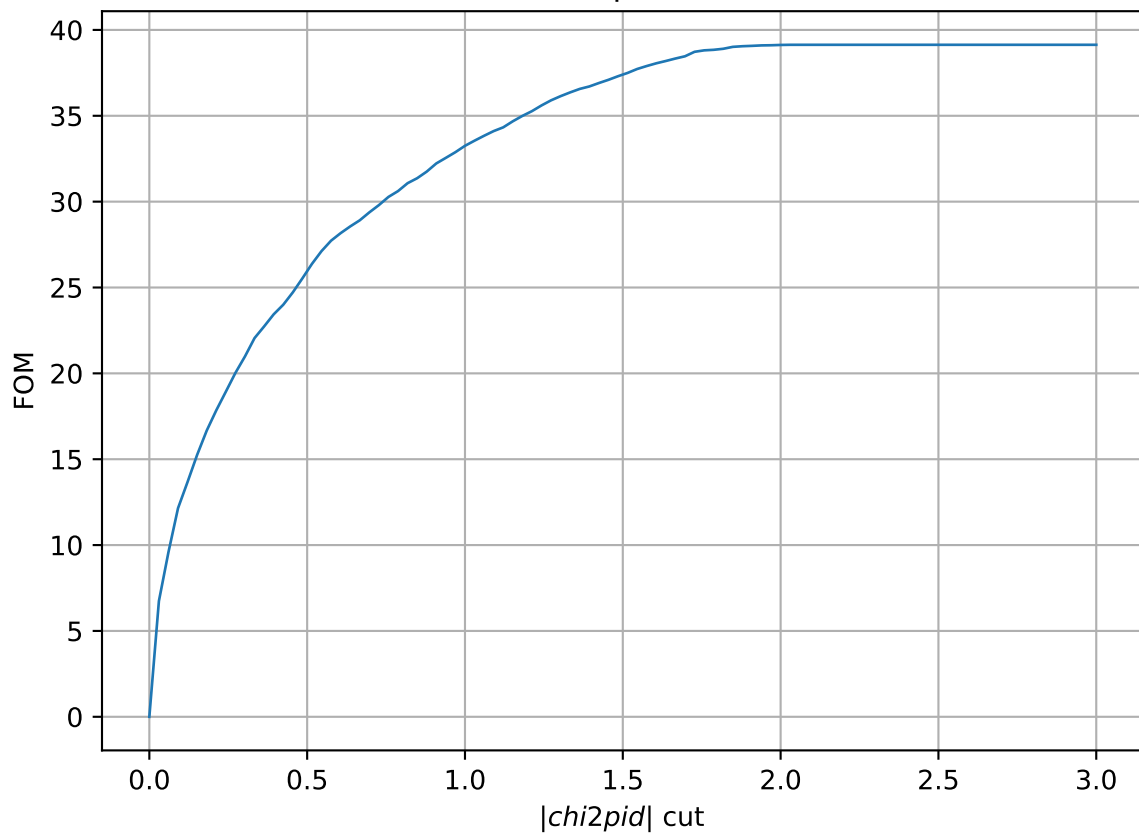
FOM vs $|\chi^2_{pid}|$ cut
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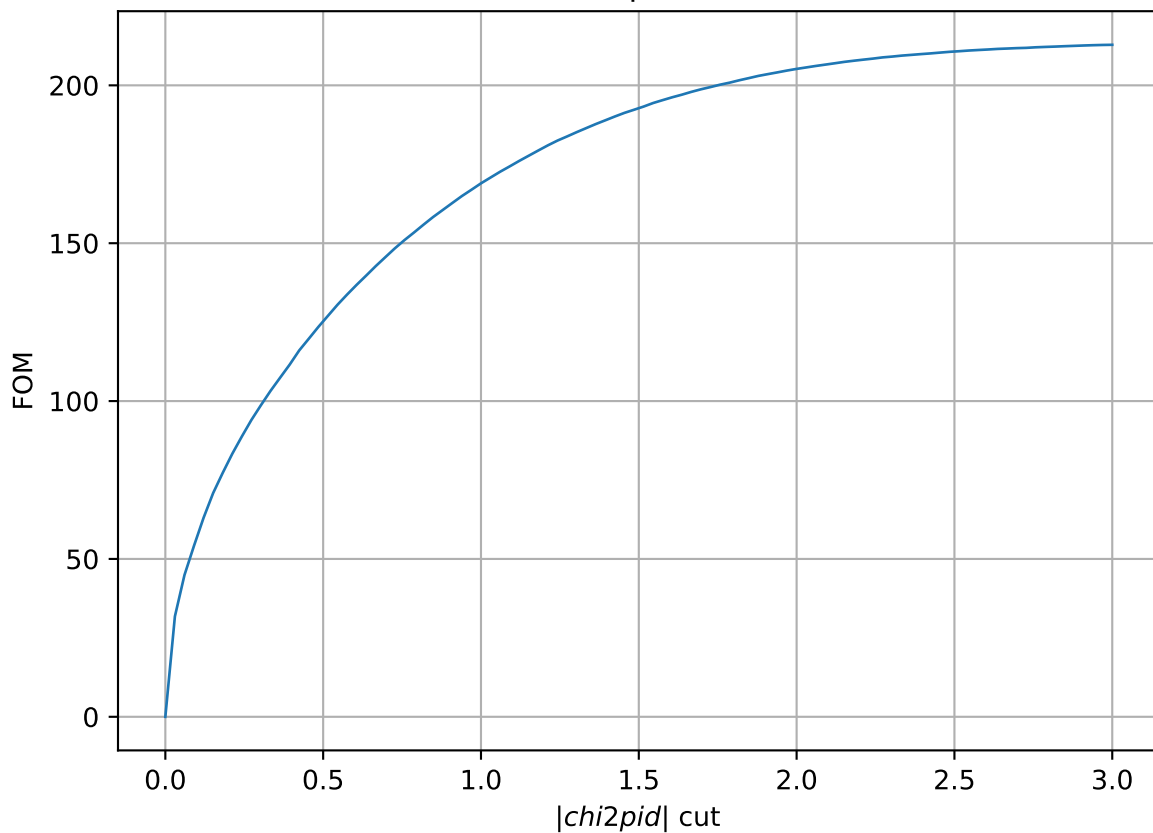
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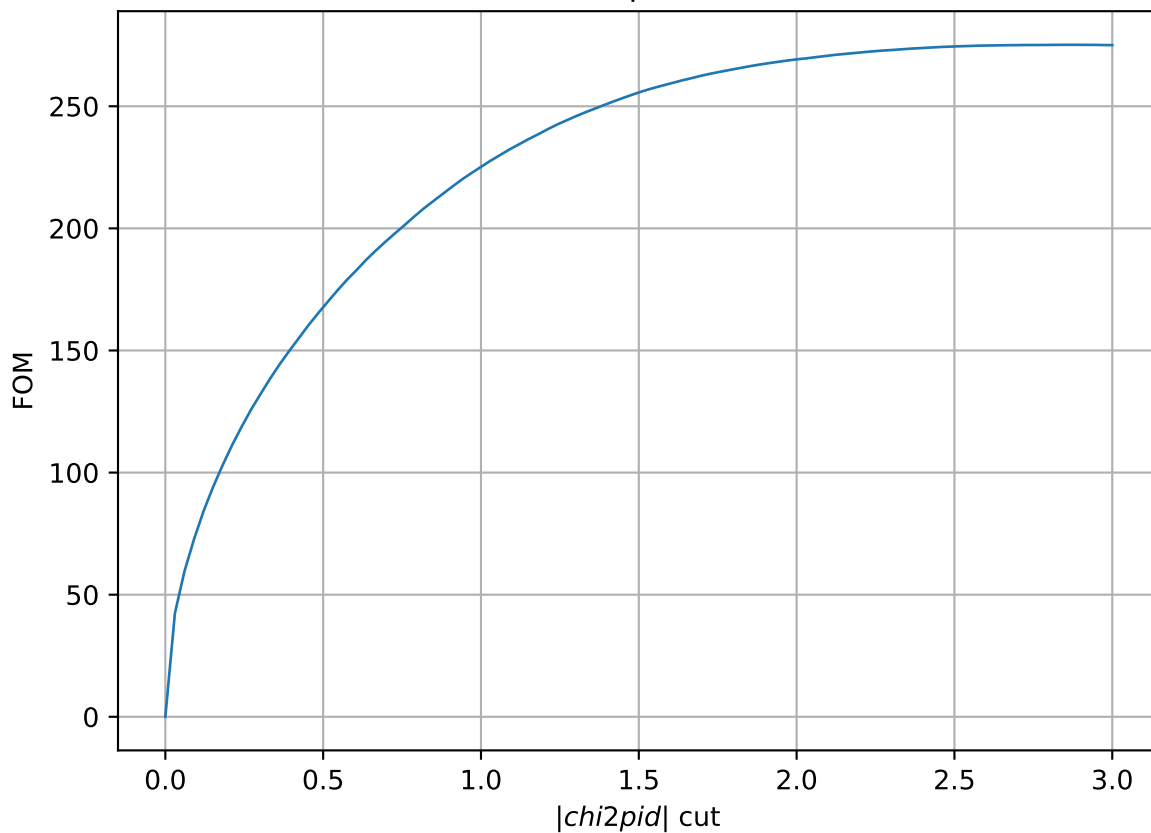
FOM vs $|\chi^2_{pid}|$ cut
 $17.0 < \theta < 23.0$
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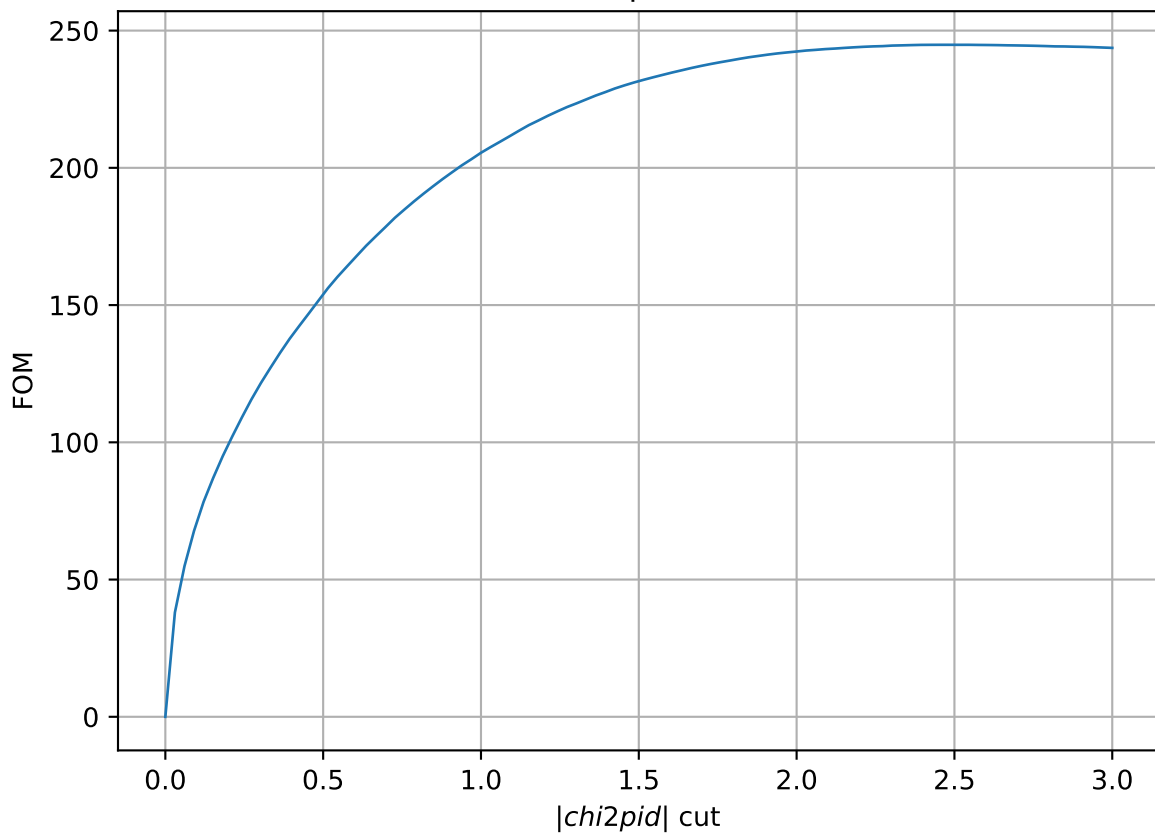
FOM vs $|\chi^2_{pid}|$ cut
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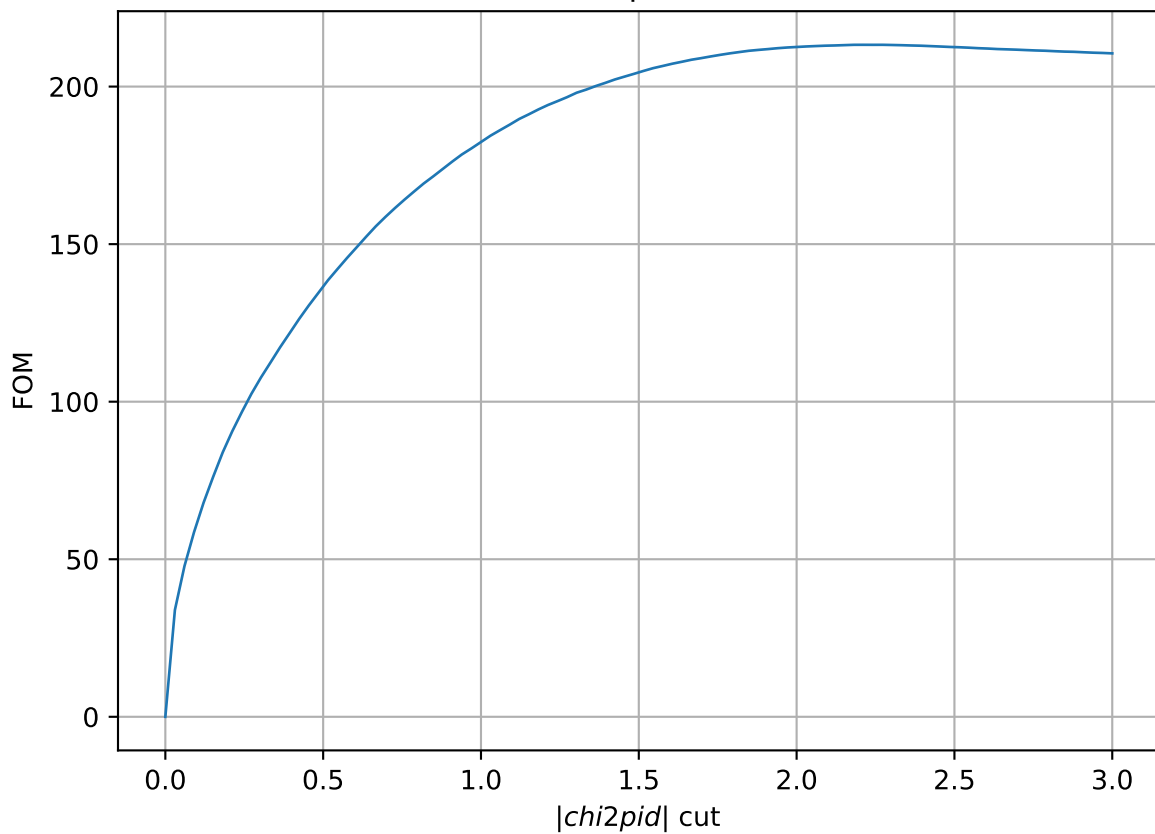
FOM vs $|\chi^2_{pid}|$ cut
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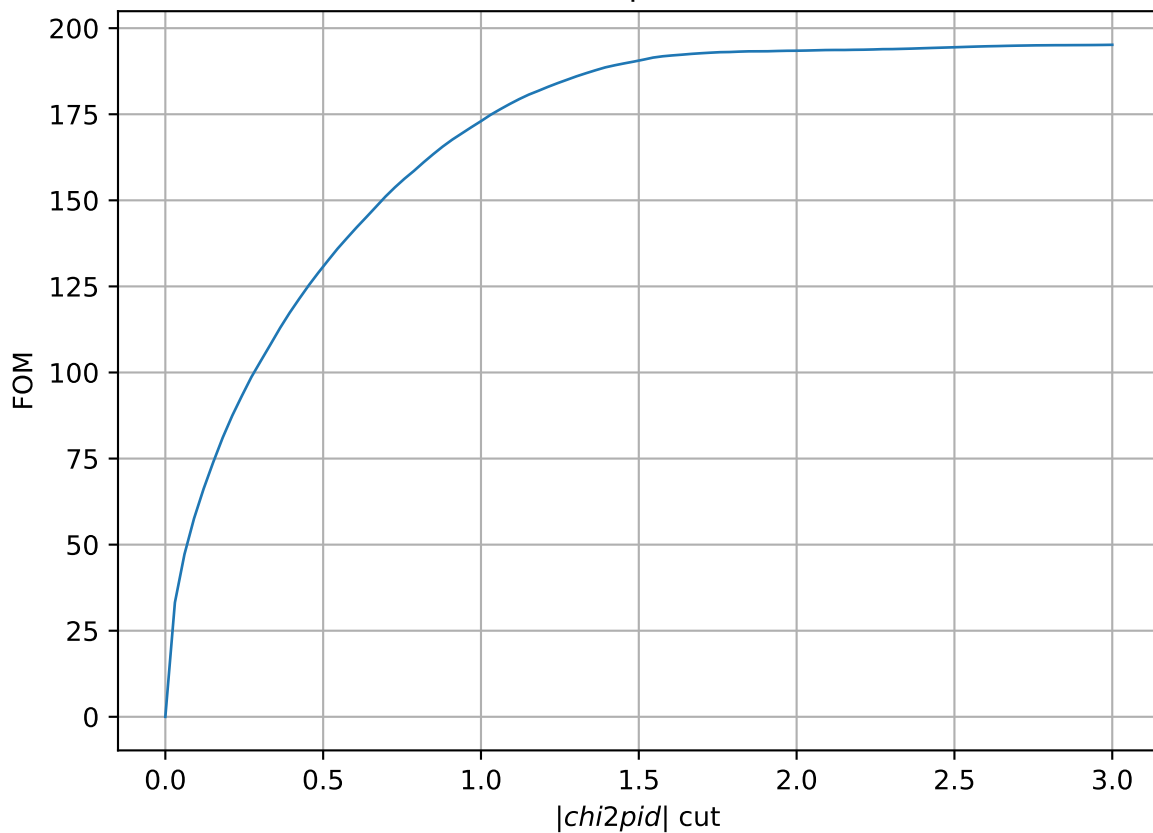
FOM vs $|\chi^2_{pid}|$ cut
 $23.0 < \theta < 29.0$
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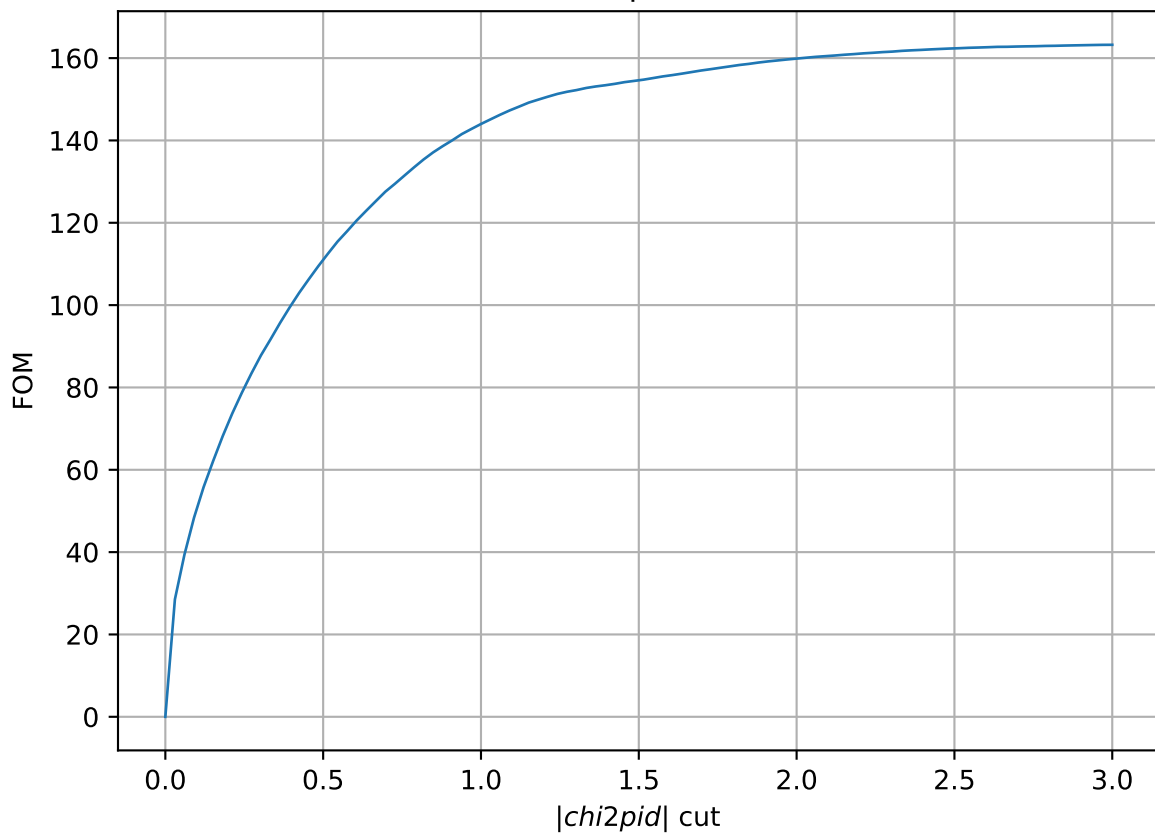
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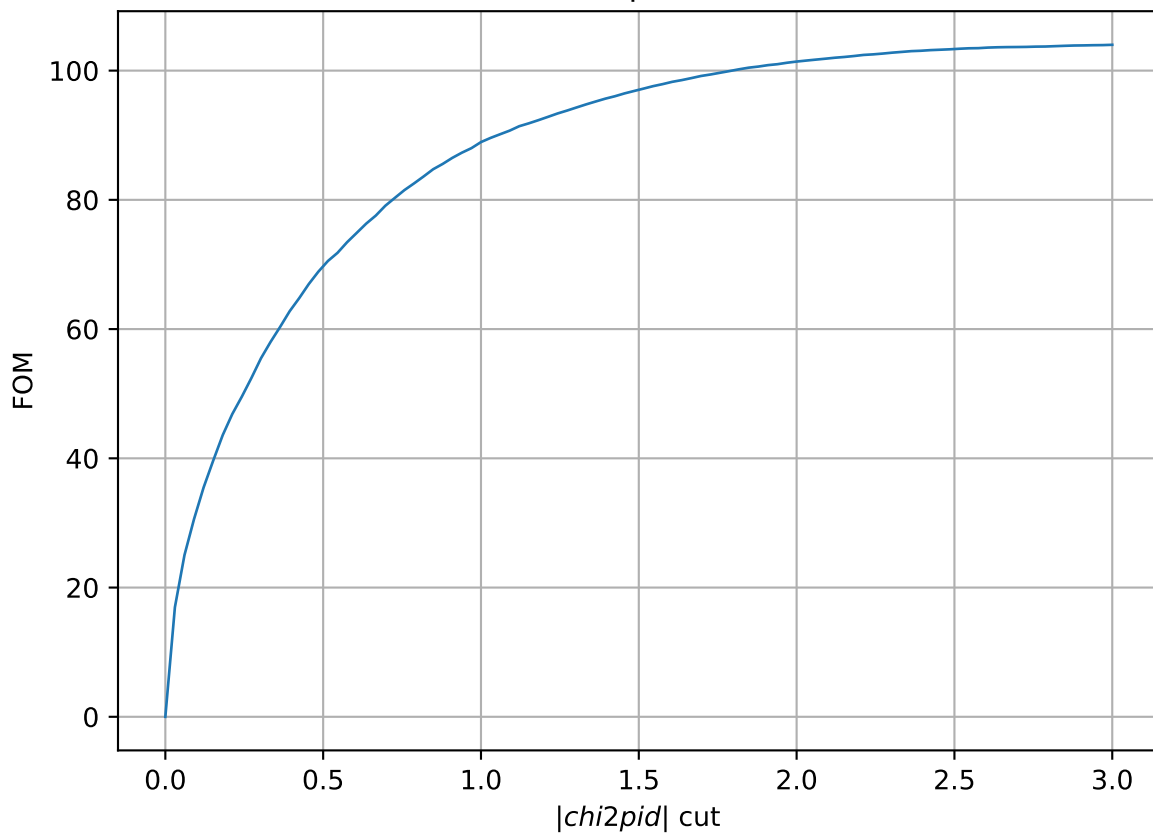
FOM vs $|\chi^2_{pid}|$ cut
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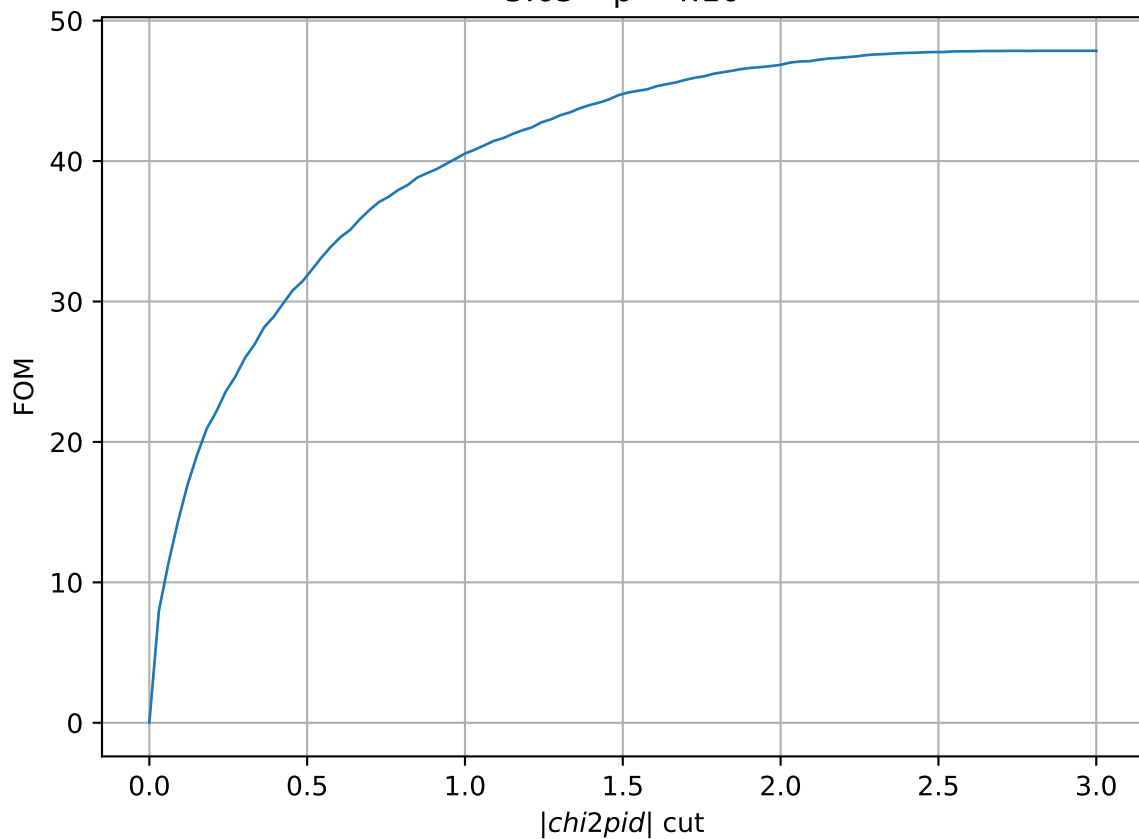
FOM vs $|\chi^2_{pid}|$ cut
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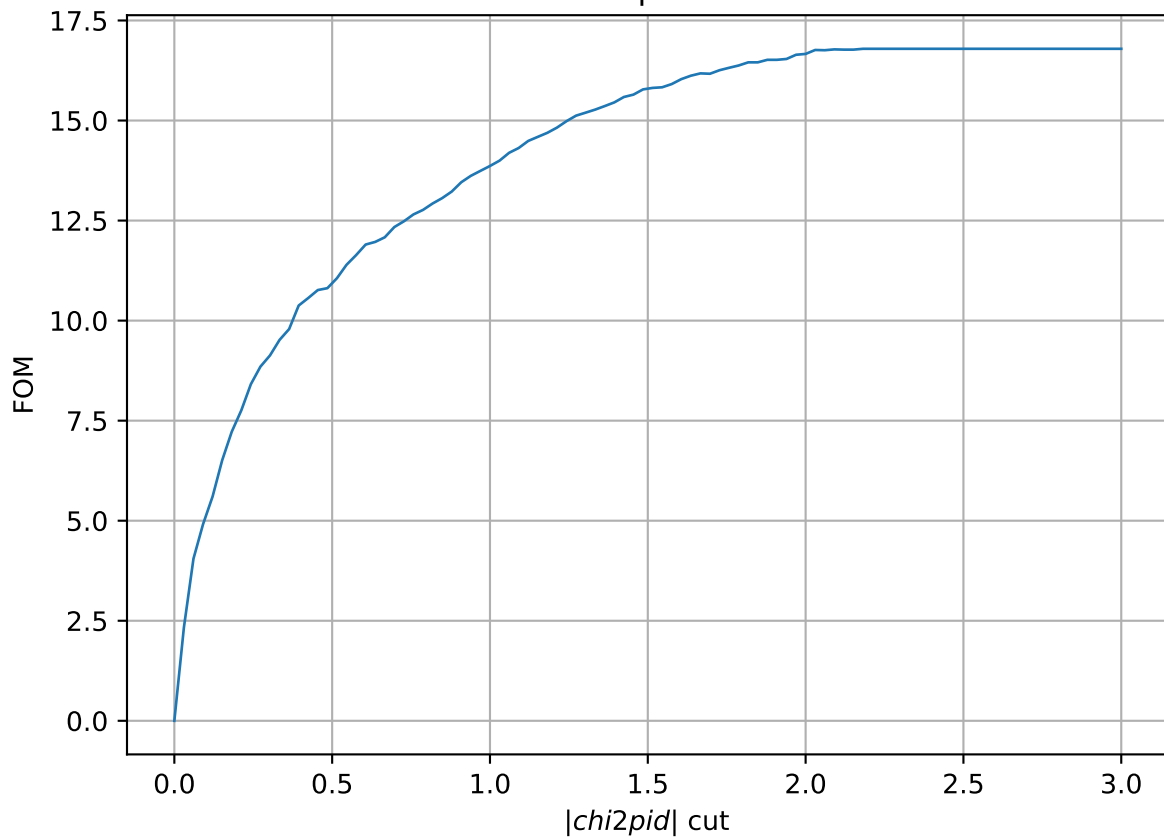
FOM vs $|\chi^2_{pid}|$ cut
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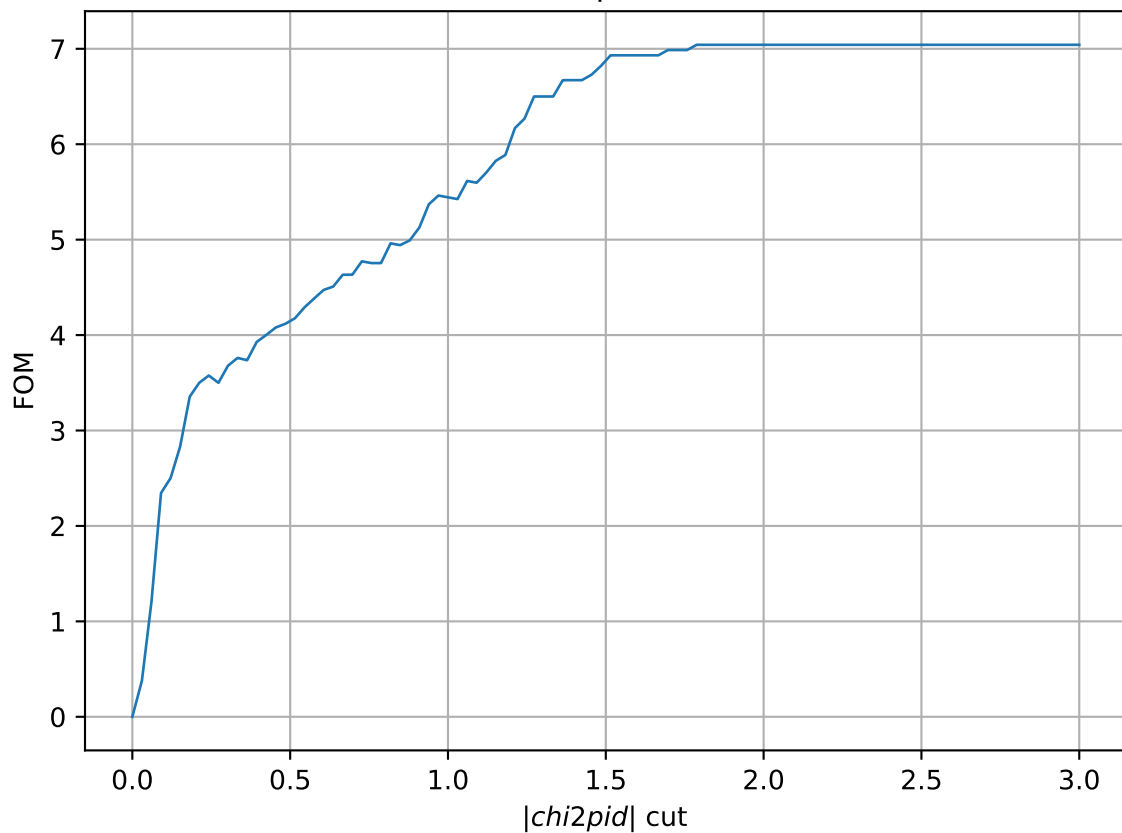
FOM vs $|\chi^2_{pid}|$ cut
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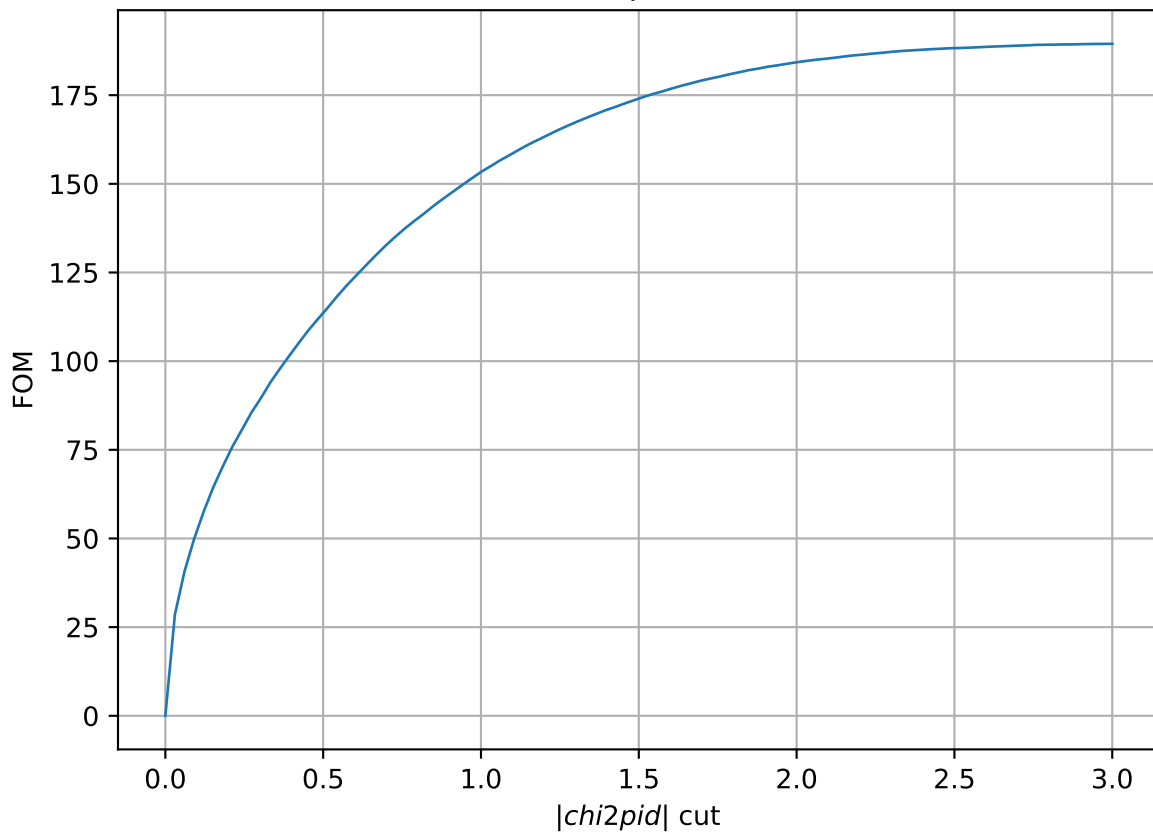
FOM vs $|\chi^2_{pid}|$ cut
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 $4.10 < p < 4.55$



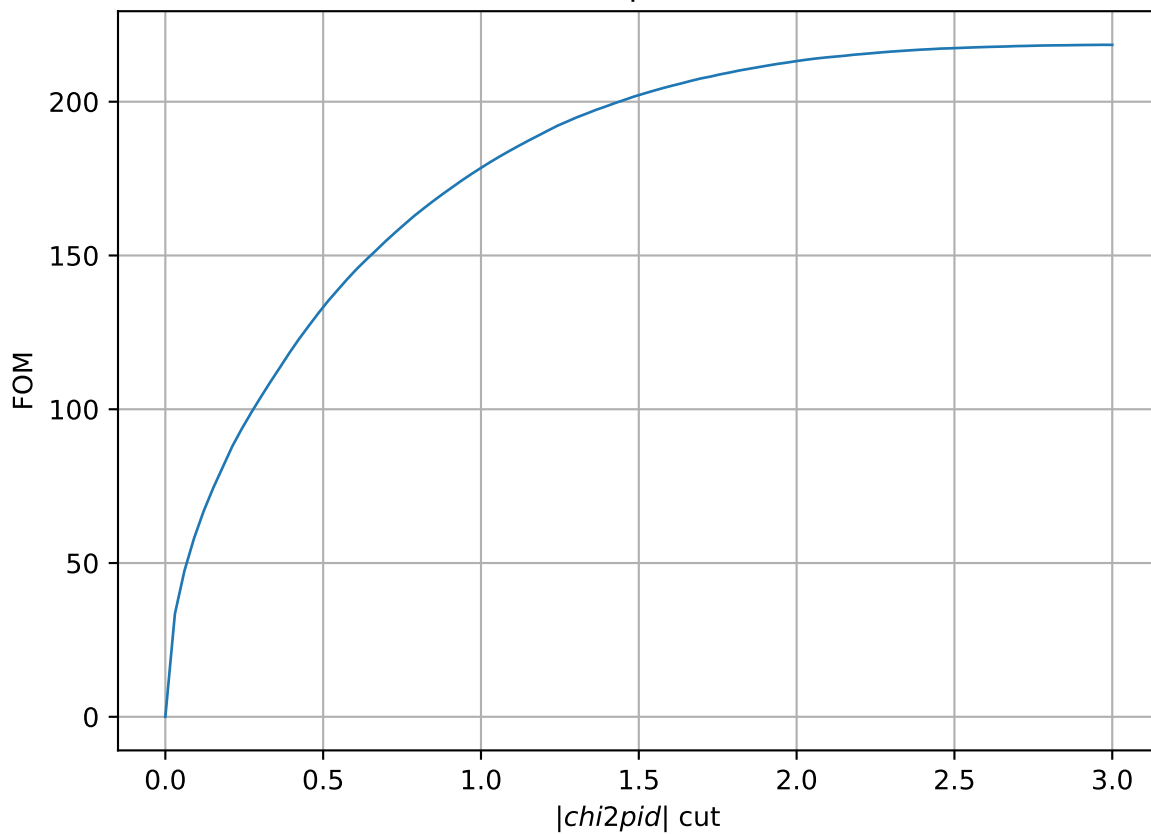
FOM vs $|\chi^2_{pid}|$ cut
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 $4.55 < p < 5.00$



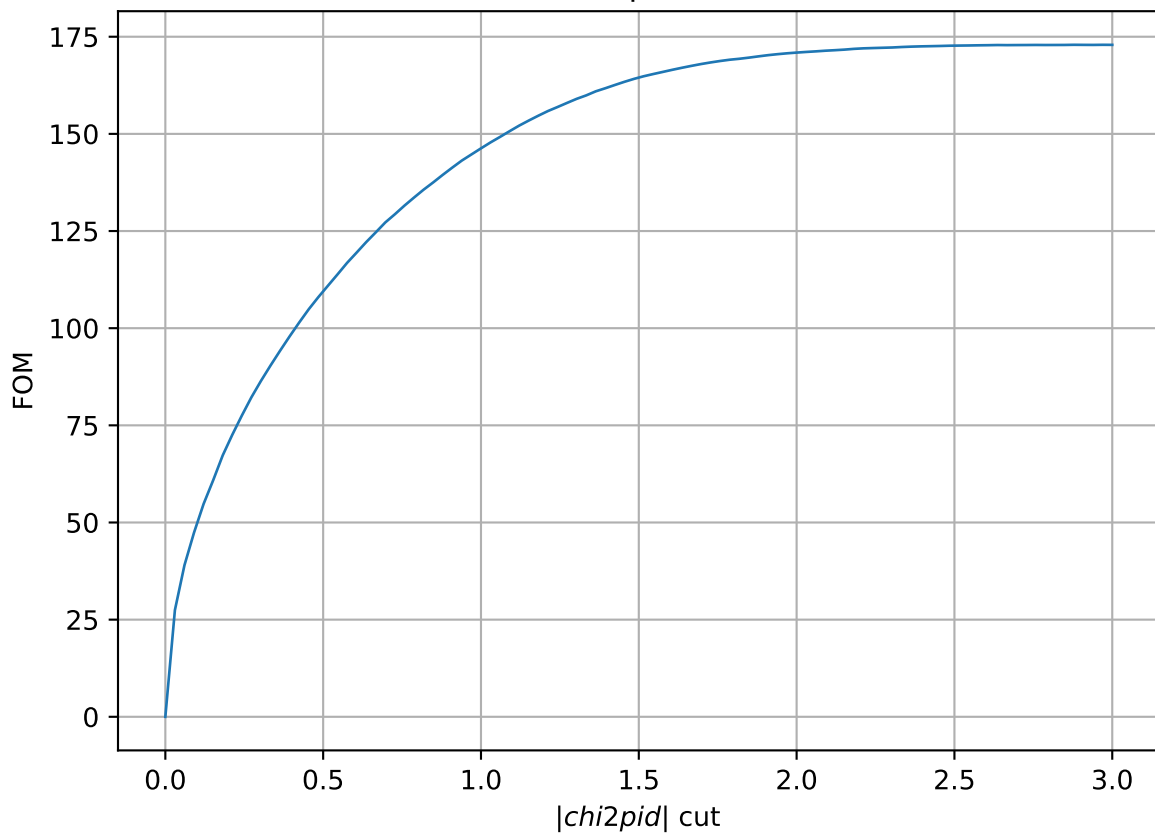
FOM vs $|\chi^2_{pid}|$ cut
 $29.0 < \theta < 35.0$
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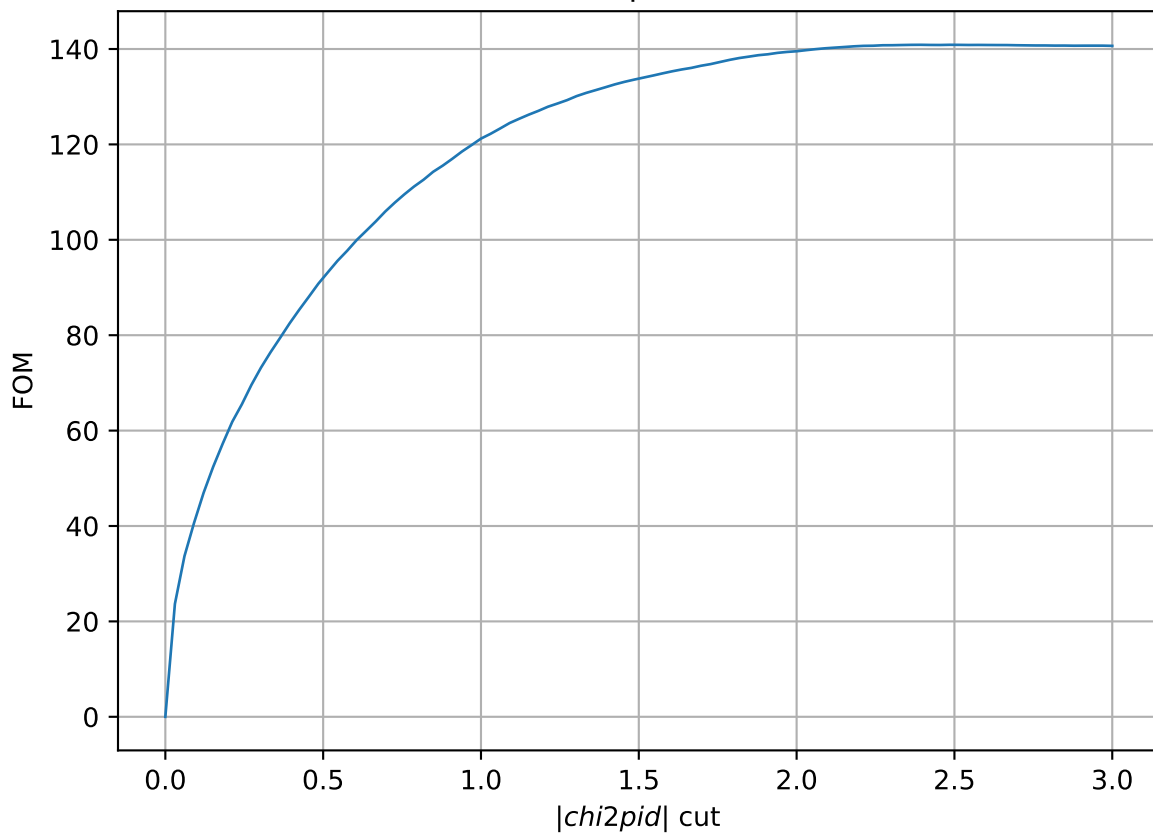
FOM vs $|\chi^2_{pid}|$ cut
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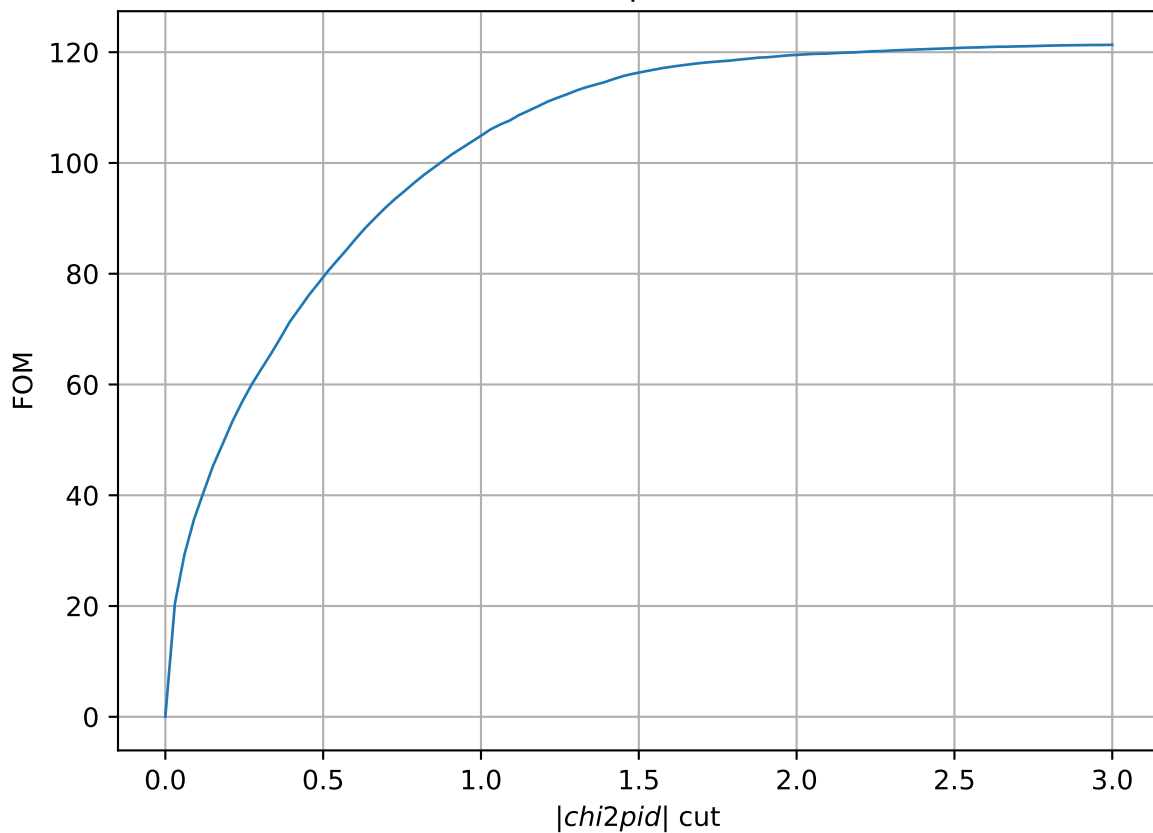
FOM vs $|\chi^2_{pid}|$ cut
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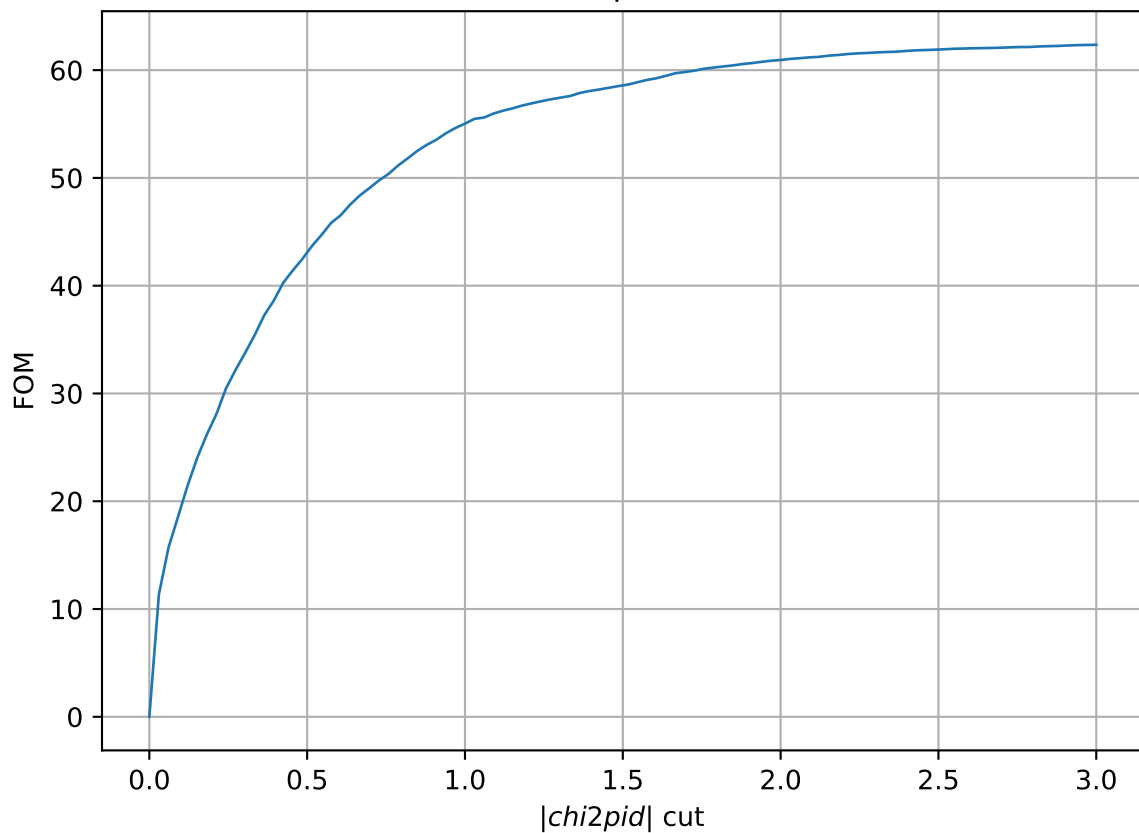
FOM vs $|\chi^2_{pid}|$ cut
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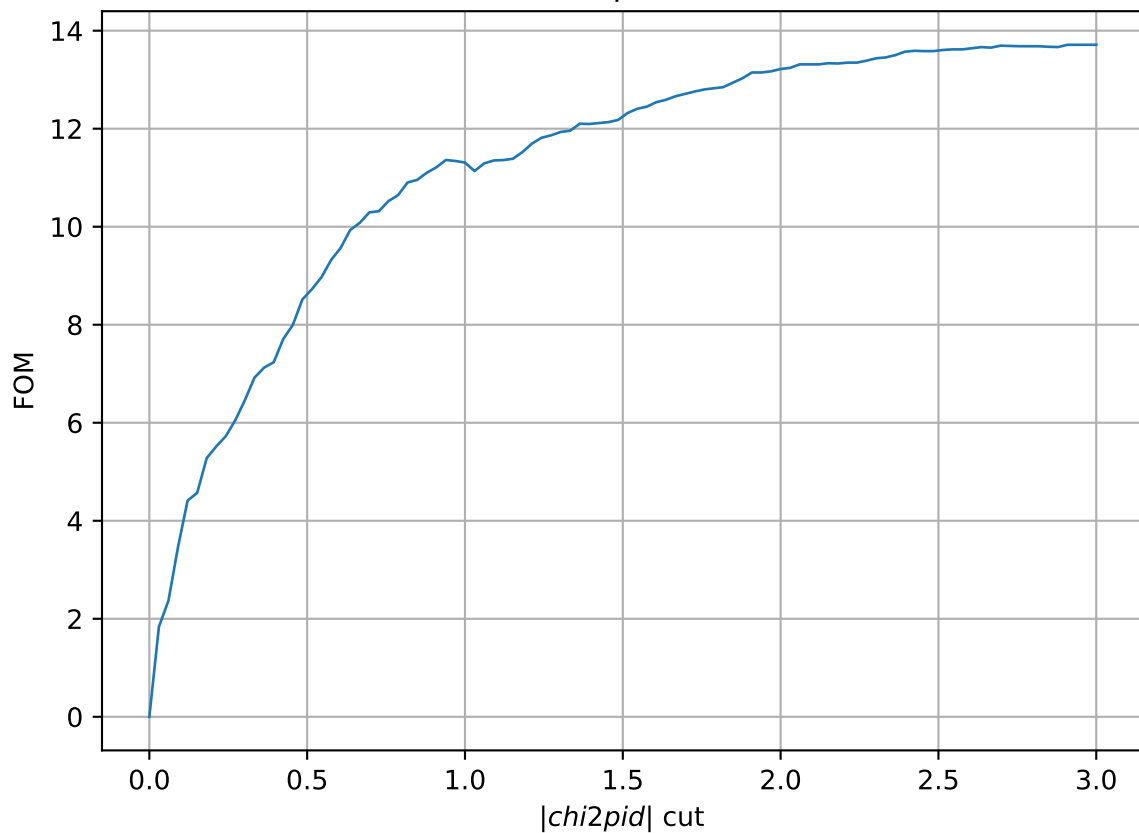
FOM vs $|\chi^2_{pid}|$ cut
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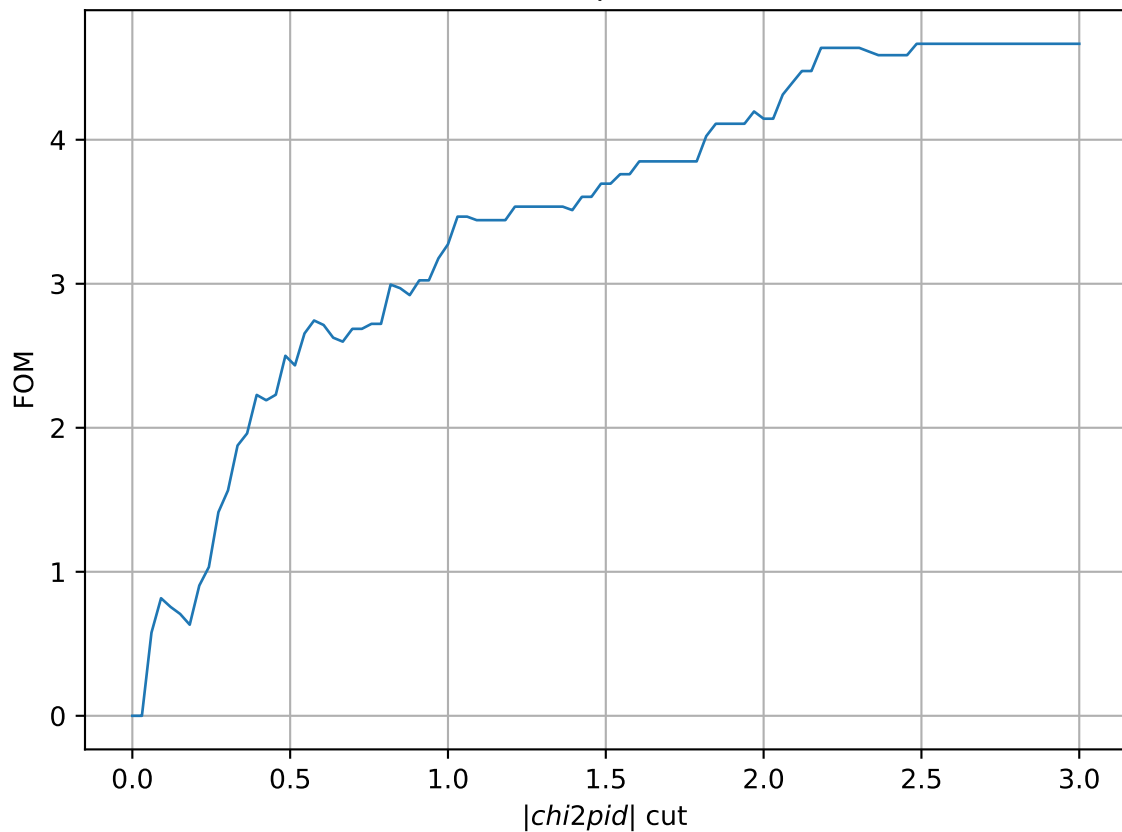
FOM vs $|\chi^2_{pid}|$ cut
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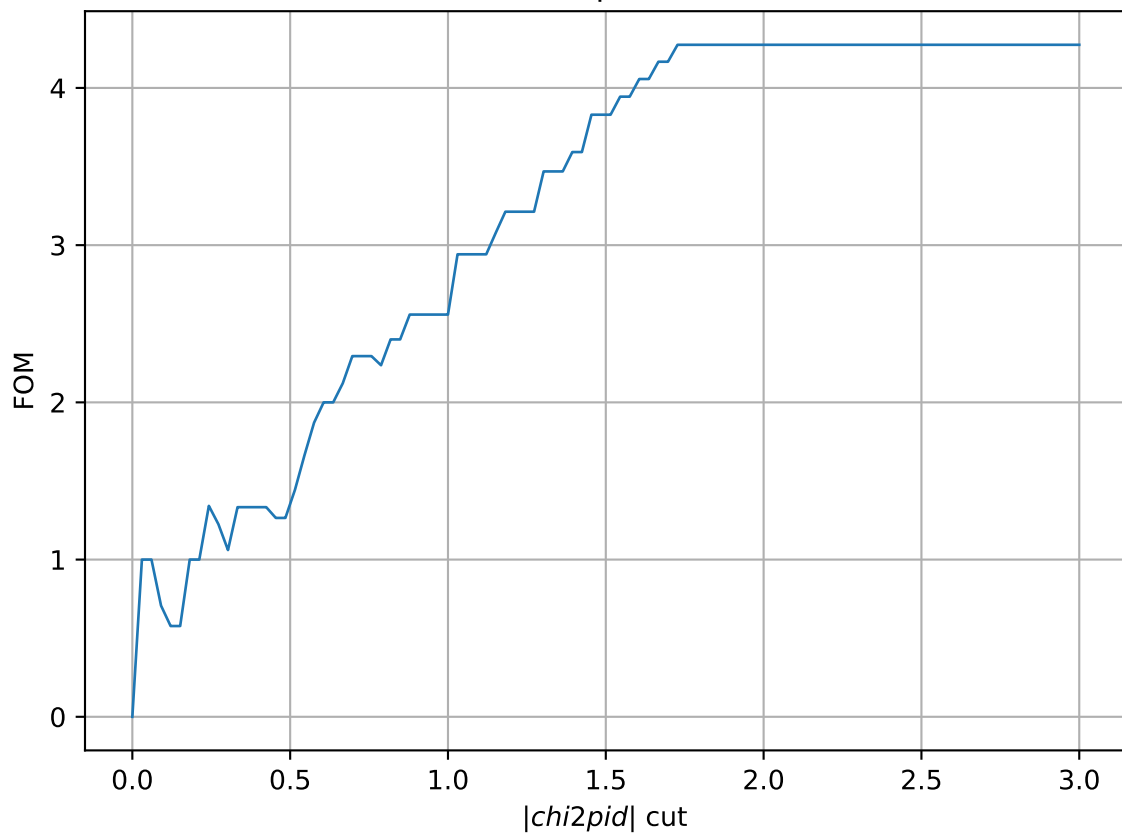
FOM vs $|\chi^2_{pid}|$ cut
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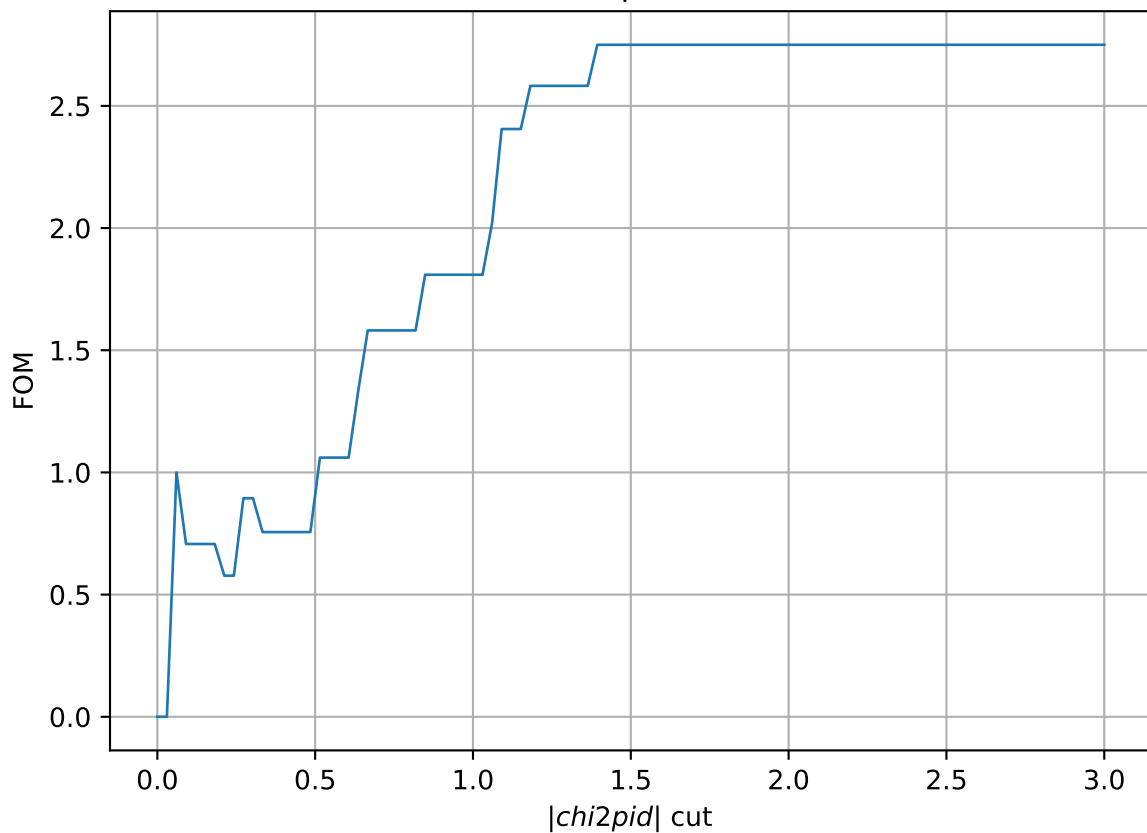
FOM vs $|\chi^2_{pid}|$ cut
 $29.0 < \theta < 35.0$
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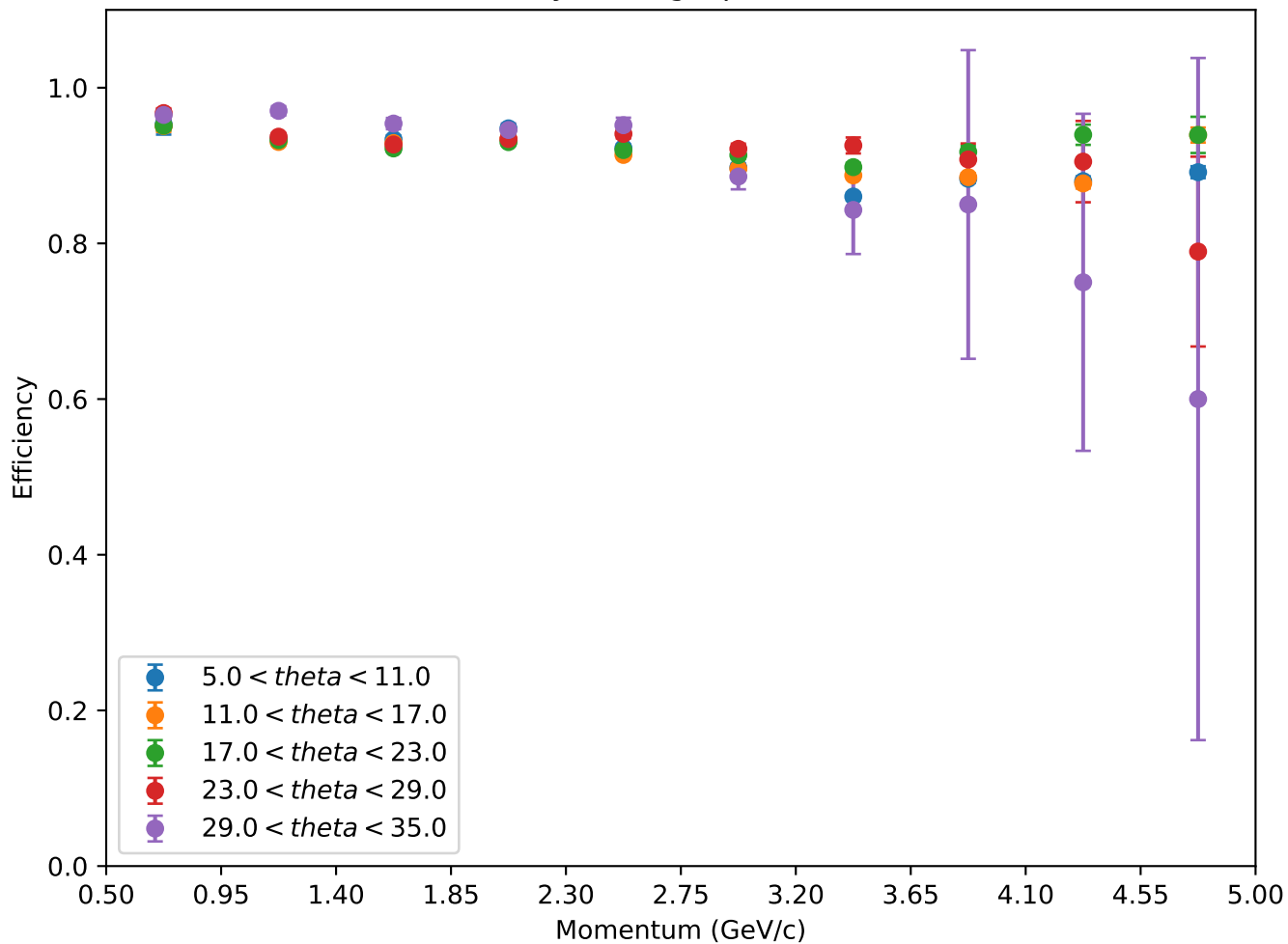
FOM vs $|\chi^2_{pid}|$ cut
 $29.0 < \theta < 35.0$
 $4.10 < p < 4.55$



FOM vs $|\chi^2_{pid}|$ cut
 $29.0 < \theta < 35.0$
 $4.55 < p < 5.00$



Efficiency (Using Optimized BDT)



Contamination (Using Optimized BDT)

